The Holland Atom II: Atomic Radius Variation and Shell-Band Geometry

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Selection of the Holland Atom reference radius and evidence for structured shell envelopes

Abstract

Atomic radius lacks a universal definition, leading to multiple coexisting systems (empirical, covalent, metallic, van der Waals, calculated) that assign markedly different values to the same element. Conventionally treated as methodological artefacts, this variation is here audited systematically across the periodic table as a potential physical observable.

Comprehensive analysis of major radius datasets reveals highly structured behaviour: reproducible hierarchical scaling, strong group and block dependence, and statistically distinct populations separating s/p-blocks from d/f-blocks. Most strikingly, shell-band amplitude (half-range of radius variation) is effectively independent of shell-band midpoint ($r = -0.028$, $p = 0.785$), demonstrating that radius spread is not merely a trivial consequence of atomic size. Robustness tests confirm an outer interaction envelope (dominated by van der Waals radii) and a structured internal radius family that persists after vdW removal. Mean observed shell-band width is ~75 pm (amplitude ~38 pm).

These findings support the selection of empirical atomic radius as the standard geometric reference for the Holland Atom programme. Within the resonance-locked standing-wave framework established in Holland Atom Paper 1 [31], the results indicate that atomic charge shells possess finite radial extent – consistent with nested standing modes in the scalar Holland Field Φ of General Expanse Tension Theory (GETT). Multiple accepted radius definitions thus appear as complementary samplings of different regions within physically extended shell envelopes rather than conflicting measurements.

This work establishes atomic radius variation as a genuine structural signal, provides the geometric reference framework for subsequent Holland Atom studies (including charge density and scaling in Paper 3), and offers empirical motivation for a deeper causal ontology of atomic matter beneath quantum-mechanical formalism.

Keywords: Atomic radius variation, shell-band structure, standing-wave resonance, Holland Atom, GETT, empirical atomic radius, periodic table, finite shell extent, resonance-lock

1. Introduction

Atomic radius is one of the most fundamental quantities in chemistry and atomic physics. It underpins the interpretation of atomic structure, periodic behaviour, bonding, crystal packing, intermolecular interactions, charge density, and countless derived properties throughout the physical sciences. Despite its importance, there is no universally accepted definition of atomic radius. Unlike macroscopic objects, atoms possess no directly observable physical surface. Consequently, atomic size is not measured directly but inferred from a variety of experimental observations, theoretical calculations, and operational conventions. Numerous radius definitions have therefore emerged,

including empirical atomic radius, covalent radius, metallic radius, ionic radius, calculated atomic radius, and van der Waals radius. These radius definitions frequently assign substantially different values to the same element. For over a century this multiplicity of accepted radius values has generally been regarded as a consequence of differing measurement methodologies and differing physical interpretations of atomic size. Under this conventional view, the variation between radius definitions is largely treated as an unavoidable limitation arising from the absence of a uniquely defined atomic boundary.

However, the existence of systematic relationships between radius systems raises an important question: **Are the various radius definitions simply alternative conventions for describing atomic size, or does the variation itself contain physical information regarding atomic structure?**

The present work investigates this question directly. The study forms part of the wider Holland Atom research programme. Holland Atom Paper 1 introduced a resonance-lock reconstruction of atomic structure and demonstrated that a standing-mode shell framework can reproduce a broad range of observed atomic behaviours, including shell ordering, shell capacities, shell filling sequences, shell contraction, anomalous configurations, and periodic-table organisation [31]. Before quantitative shell-scaling laws can be developed, a standard geometric radius must first be established. The *original* purpose of this paper was therefore to determine which radius definition should be adopted as the reference radius for future Holland Atom studies. During the course of that investigation, a broader and unexpected observation emerged. The variation between accepted radius definitions itself **appears to exhibit non-trivial systematic structure throughout the periodic table**.

This paper therefore pursues two closely related objectives.

First, it performs a comprehensive audit of the principal atomic radius systems and determines which radius definition provides the most suitable geometric reference for future Holland Atom development.

Second, it tests a specific prediction arising from Holland Atom Paper 1: if atomic charge shells are resonance-locked standing modes, then the shell should possess finite radial extent rather than an infinitesimally thin boundary. **Under this hypothesis, the observed spread between accepted atomic radius definitions may contain measurable information regarding shell depth, shell-band structure, and the amplitude of the underlying standing mode.** The present study therefore investigates whether atomic radius variation exhibits the statistical organisation and physical characteristics expected of such a shell structure. To achieve this, the study proceeds through a series of progressively larger analyses.

- **Section 2** establishes the observational basis of atomic radius variation and demonstrates the phenomenon using representative examples – the current world view empirical baseline.
- **Section 3** extends the analysis to a full periodic-table audit of the principal radius systems.
- **Section 4** investigates whether the observed radius variation exhibits statistically significant periodic structure.
- **Section 5** introduces the Holland Atom interpretation and explores the possibility that multiple radius definitions may probe different regions within a finite shell structure.
- **Section 6** determines the standard geometric radius to be adopted throughout the Holland Atom research programme.
- **Section 7** quantifies the observable properties of the resulting shell-band structure.
- **Section 8** discusses the significance of the findings, compares them with existing understanding, and evaluates their implications for atomic structure.

The central proposition examined throughout this work is that the variation between accepted atomic radius definitions may reflect the finite radial extent of atomic charge shells. If so, radius variation

should exhibit systematic organisation throughout the periodic table rather than behaving as random measurement disagreement. The analyses that follow demonstrate that this is indeed the case.

2. The Observed Phenomenon of Atomic Radius Variation

2.1 Introduction

Despite its widespread use, there is no universally accepted physical boundary that uniquely defines the size of an atom. Instead, multiple radius definitions have been developed for different experimental, theoretical, and practical applications. These include empirical atomic radii, covalent radii, ionic radii, metallic radii, van der Waals radii, electron-density radii, and various theoretically derived quantum-mechanical radii. Importantly, these radius definitions are not generally equivalent. The same element may be assigned substantially different radius values depending upon the measurement methodology employed. The existence of multiple accepted radius values for the same atom represents a fundamental observational fact of modern chemistry and atomic physics.

This section establishes the empirical baseline for that phenomenon.

2.2 There Is No Universally Accepted Atomic Boundary

Unlike macroscopic objects, atoms do not possess a directly observable physical surface. Consequently, atomic size cannot be measured in the same manner as the diameter of a solid object. Instead, atomic radius is inferred from a variety of experimental observations and theoretical constructs, including:

- bond lengths
- crystal structures
- intermolecular separations
- electron-density calculations
- wavefunction properties
- theoretical scale parameters

The resulting radius definitions therefore represent different operational descriptions of atomic size rather than direct measurements of a single physical boundary. Appendix A provides a comprehensive catalogue of the principal radius definitions used throughout chemistry and physics.

Observation 1: ***No commonly used atomic radius is directly measured as a physical radius.***

All accepted atomic radii are inferred, assigned, derived, or calculated from other experimentally observed quantities or theoretical models.

2.3 Multiple Radius Definitions Exist

Appendix A identifies fourteen commonly used radius definitions. These include:

Bonding-Based Radii

- Covalent Radius [12-15]

- Single-Bond Covalent Radius
- Double-Bond Covalent Radius
- Triple-Bond Covalent Radius
- Empirical Covalent Radius [12-15]

Structure-Based Radii

- Empirical Atomic Radius [9,10]
- Metallic Radius [18]
- Ionic Radius [11]
- van der Waals Radius [8,16]

Quantum-Theoretical Radii

- Calculated Atomic Radius [9-10]
- Expectation Radius
- RMS Radius
- Electron-Density Contour Radius

Atomic-Theory Scale Parameters

- Bohr Radius [1]

Each radius definition is associated with a distinct measurement methodology and physical interpretation.

Observation 2: ***Atomic radius is not a uniquely defined atomic parameter.***

Rather, atomic radius represents a family of related measurements that describe different aspects of atomic structure and behaviour.

Radius Type	Symbol	Basic Definition	Typical Method of Determination
Atomic Radius (Empirical)	Ra	Effective atomic size derived from observed atomic dimensions and periodic trends.	Experimental observations of crystal structures, densities, and atomic packing.
Atomic Radius (Calculated)	$Ra{-}c$	Atomic radius predicted from theoretical electron-density calculations.	Quantum-mechanical electron-density modelling.
Covalent Radius (2008)	R_{c2008}	Half the distance between two identical bonded atoms.	Crystallographic bond-length analysis (Cordero et al., 2008).
Covalent Radius (Single Bond)	$R_{c,s}$	Radius associated with a typical single covalent bond.	Half measured single-bond length.
Covalent Radius (Triple Bond)	$R_{c,t}$	Radius associated with a typical triple covalent bond.	Half measured triple-bond length.
Metallic Radius	R_m	Effective radius of atoms within metallic crystal structures.	Half nearest-neighbour separation in metallic lattices.
van der Waals Radius	R_{vdW}	Outer non-bonded interaction radius of an atom.	Half closest approach distance between non-bonded atoms.

Table 1. Principal Atomic Radius Families Used in this Study

Table 1 shows the radius families used throughout this study span experimental, theoretical, bonding, crystalline, and interaction-derived measures of atomic size [8,16]. *Although each radius system is widely accepted within its respective domain, they frequently assign substantially different values to the same element. The present work investigates whether these differences represent measurement convention alone or whether they contain information regarding underlying atomic shell structure.*

The coexistence of multiple radius definitions has long been recognised within chemistry, where different radius systems are adopted according to bonding environment, crystal structure, and atomic interaction regime [7].

2.4 Carbon Demonstrates Radius Variation Within a Single Element

To illustrate the phenomenon directly, Appendix B examines Carbon as a representative case study. Carbon was selected because it possesses one of the most complete collections of radius measurements available for any element. The Carbon dataset includes:

Radius Type	Radius (pm)
Inner Radial Node	~30
Triple-Bond Covalent Radius	60
Double-Bond Covalent Radius	67
Calculated Atomic Radius	67
Empirical Atomic Radius	70
Covalent Radius (2008)	76
Single-Bond Covalent Radius	76
Empirical Covalent Radius	77
van der Waals Radius	170

Table 2. Radius Variation in Carbon across Radius Systems

Table 2 shows the accepted radius systems for Carbon span approximately 60–170 pm, corresponding to a range of almost a factor of three. This demonstrates that a single element may legitimately possess multiple accepted atomic radius values depending upon the measurement or calculation methodology employed.

In addition, the radial probability distribution contains an inner radial node at approximately 30 pm. Although this quantity is not treated as an atomic radius measurement and is therefore excluded from the accepted radius family, it provides a useful structural indicator of the internal shell geometry. Inclusion of this inner structural boundary extends the observable radial span from approximately 30 pm to 170 pm, corresponding to a factor of approximately 5.7.

The Carbon case study does not compare multiple measurements of an identical physical state. Rather, it compares accepted radius definitions derived from different experimental, theoretical, bonding, and interaction contexts. The objective is not to establish a single correct Carbon radius, but to examine whether these accepted radius systems exhibit a consistent and interpretable hierarchy.

Observation 3: ***Radius variation exists within a single element.***

The phenomenon is therefore not a consequence of comparing different atoms but is present even when examining a single atomic species.

2.5 Radius Variation Is Systematic

Appendix B Part 2 demonstrates that the Carbon radius values are not randomly distributed. Several systematic relationships emerge.

Bond Order Dependence

Within the covalent-radius family:

Bond Type	Radius (pm)
Single Bond	76
Double Bond	67
Triple Bond	60

Table 3. Carbon Bond Order Dependence

Table 3 shows that as bond order increases: Single → Double → Triple the assigned radius decreases. This behaviour follows directly from experimentally observed reductions in bond length.

Clustering Behaviour

Calculated, empirical, and covalent radius definitions cluster within a relatively narrow interval: 67–77 pm, despite being derived from different methodologies.

Outer-Boundary Behaviour

The van der Waals radius is substantially larger than all other neutral-atom radius definitions examined.

Observation 4: ***Radius variation exhibits structure and ordering rather than random scatter.***

2.6 Radius Availability Depends Upon Measurement Methodology

Appendix C further demonstrates that not all radius definitions are available for all elements. Examples include:

- missing empirical radii for certain noble gases
- absence of metallic radii for non-metallic elements
- multiple competing ionic-radius assignments
- differing availability of bond-order-specific radii

Observation 5: ***The existence of a radius value depends upon the measurement methodology employed.***

Consequently, radius definitions are properties of both the atom and the measurement framework used to characterise it.

2.7 Comparative Analysis of Radius Systems

Appendix C compares several major radius systems across a representative multi-element dataset. The analysis demonstrates that:

- Covalent Radius and Empirical Covalent Radius closely track Empirical Atomic Radius.
- van der Waals Radius is consistently larger than Empirical Atomic Radius.
- Radius systems exhibit reproducible relative ordering across chemically diverse elements.
- Relative ratios are often more stable than absolute radius values.

These observations suggest that the principal radius systems may be related through a common structural hierarchy.

Observation 6: ***Different radius definitions exhibit reproducible relationships across multiple elements.***

2.8 Sample Element Comparative Analysis

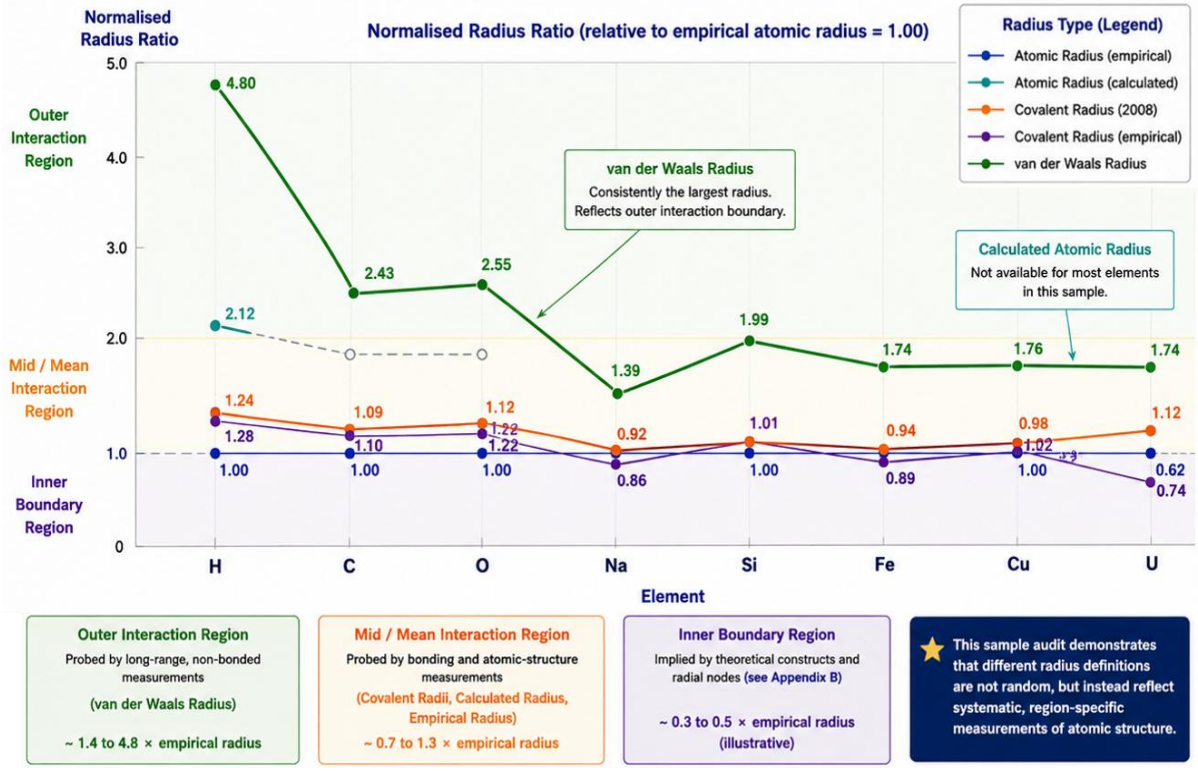


Figure 1. Normalised Radius Ratios Relative to Empirical Atomic Radius

Figure 1 presents the normalised radius ratios obtained from the representative multi-element sample described in Appendix C. Each radius system is expressed relative to the Empirical Atomic Radius (R_a), which is assigned a value of 1.00 for each element. The sample includes hydrogen, carbon,

oxygen, sodium, silicon, iron, copper, and uranium, representing a diverse cross-section of chemical families and periodic-table regions [19-20].

Several important observations emerge.

First, the principal radius systems exhibit a reproducible ordering across chemically diverse elements. The Empirical Atomic Radius remains fixed at unity by definition, while both Covalent Radius (2008) and Empirical Covalent Radius remain closely clustered around this value. Despite being derived from different methodologies, these radius systems consistently track the empirical atomic radius, typically within approximately ±25%.

Second, the van der Waals Radius forms a clearly distinct population. For every element examined, the van der Waals radius exceeds the empirical atomic radius and remains the largest neutral-atom radius definition in the dataset. The observed ratios range from approximately 1.4 to 4.8 times the empirical atomic radius, suggesting that van der Waals measurements probe a substantially more external interaction region than bonding-based or structure-based radius definitions.

Third, although absolute radius values vary considerably from element to element, the normalised ratios exhibit substantially greater stability. This indicates that the relationships between radius systems may be more fundamental than the individual radius values themselves.

The figure also reveals the emergence of a potential hierarchical organisation of radius measurements. Covalent, calculated, and empirical radius systems cluster within a relatively narrow intermediate region, whereas van der Waals radii occupy a distinct outer region. The illustrative inner-boundary region identified in the Carbon case study (Appendix B) lies below these measurements and is not directly represented by the principal radius systems commonly used throughout chemistry.

Importantly, the observed ordering is not random. The same relative hierarchy appears repeatedly across the representative sample despite substantial differences in atomic number, electron configuration, bonding behaviour, and chemical classification. **This reproducibility suggests that the various radius definitions may be probing different radial domains of atomic structure rather than representing unrelated measures of atomic size.**

The purpose of Figure 1 is therefore not merely to compare numerical radius values, but to demonstrate that atomic radius variation exhibits systematic and reproducible behaviour. The existence of such ordered relationships motivates the search for a physical explanation capable of accounting for why multiple accepted radius definitions exist and why they consistently occupy distinct relative positions within the observed radius hierarchy.

2.9 Baseline Conclusions

The observational evidence presented in Appendices A–C establishes six baseline findings.

1. There is no universally accepted atomic boundary.
2. Multiple accepted radius definitions exist.
3. Radius variation occurs within a single element.
4. Radius variation is systematic rather than random.
5. Radius availability depends upon measurement methodology.
6. Different radius systems exhibit reproducible relationships across chemically diverse elements.

Collectively, these observations establish atomic radius variation as a genuine empirical phenomenon requiring explanation.

The remainder of this paper therefore investigates whether the observed hierarchy of radius definitions can be understood as different measurements probing different regions of a common underlying atomic structure.

3. Full Periodic-Table Audit of Atomic Radius Systems

3.1 Objective

The Carbon case study (Appendix B) demonstrated that a single element may possess multiple accepted radius values depending upon the measurement or derived calculation methodology employed. Appendix C further showed that similar behaviour is observed across a representative multi-element sample. The objective of this section is to extend that investigation to the full available periodic-table dataset and determine whether the principal atomic radius systems exhibit systematic scaling relationships. The primary audit dataset comprised seven radius families: empirical atomic radius, calculated atomic radius, Cordero covalent radius (2008), single-bond covalent radius, triple-bond covalent radius, metallic radius, and van der Waals radius. These were selected to provide coverage of empirical, theoretical, bonding, crystalline, and interaction-derived atomic dimensions.

This section references Appendix D, the full periodic table audit, particularly the analysis presented in Sections 3.3 to 3.7.

The Empirical Atomic Radius was adopted as the reference standard, and all comparison datasets were normalised relative to R_a.

3.2 Dataset Coverage

The compiled master dataset contains all available radius assignments from the selected literature sources. Coverage varies significantly between radius systems. The Cordero (2008) covalent radius provides the greatest coverage, followed by empirical atomic radius and calculated atomic radius. van der Waals radius values are available for substantially fewer elements. This observation is important in its own right. No single radius system provides complete coverage of the periodic table. The availability of a radius value depends upon the methodology used to define and determine that radius. Consequently, atomic radius cannot be considered a universally measured physical property. Instead, radius values are assigned, inferred, derived, or calculated according to the measurement framework employed.

3.3 Global Scaling Relationships

The full periodic-table audit reveals clear and reproducible scaling relationships between the principal radius systems. The strongest agreement occurs between the Empirical Atomic Radius and the Cordero (2008) Covalent Radius.

The mean normalised ratio is:

$$\frac{R_{c2008}}{R_a} = 1.029$$

indicating that the modern covalent radius occupies essentially the same radial scale as the empirical atomic radius across the periodic table. The Calculated Atomic Radius exhibits a slightly larger mean ratio:

$$\frac{R_{calc}}{R_a} = 1.122$$

suggesting that electron-density-derived radii generally extend beyond the empirical radius.

The van der Waals Radius forms a distinctly larger radius family:

$$\frac{R_{vdW}}{R_a} = 1.549$$

and exhibits substantially greater variation between elements. These results demonstrate that the radius systems do not represent random or unrelated measurements. Instead, they occupy reproducible relative positions throughout the periodic table.

3.4 Correlation Between Radius Systems

Strong positive correlations are observed between the empirical atomic radius and all comparison datasets.

The strongest relationship occurs between Empirical Atomic Radius and Cordero (2008) Covalent Radius, indicating that both systems track a common underlying atomic dimension. Calculated Atomic Radius also exhibits strong correlation with Empirical Atomic Radius, although with greater dispersion. The van der Waals Radius remains correlated with atomic size but shows significantly larger variation, suggesting that it represents a distinct radial domain rather than a simple scaling of the empirical radius.

3.5 Group and Family Behaviour

The audit demonstrates that radius relationships are not uniform across the periodic table. Distinct group-dependent behaviour is observed. The largest van der Waals expansions relative to empirical radius occur within:

- Group 1 (Alkali Metals)
- Group 14 (Carbon Family)
- Group 15 (Pnictogens)
- Group 16 (Chalcogens)
- Group 17 (Halogens)

Other families, particularly many metallic and transition-metal groups, exhibit substantially smaller relative expansions. This behaviour indicates that radius variation possesses a strong family-dependent component and cannot be represented by a single universal scaling factor.

3.6 Period Behaviour

Period-dependent trends are also evident. The relative difference between van der Waals Radius and Empirical Atomic Radius is greatest among the lighter elements and decreases progressively with increasing period number. By contrast, the Cordero (2008) Covalent Radius remains remarkably stable relative to the Empirical Atomic Radius throughout the periodic table. This stability reinforces the conclusion that these two radius systems occupy closely related regions of atomic structure.

3.7 Radius Hierarchy

The periodic-table audit confirms the existence of a reproducible hierarchy of radius measurements. For every element for which paired data are available:

$$R_{vdW} > R_a$$

The van der Waals Radius therefore forms a consistently larger radius family than the Empirical Atomic Radius. The relationship between Covalent Radius and Empirical Atomic Radius is much closer, with both occupying a common intermediate scale. This behaviour mirrors the hierarchy previously identified in the Carbon case study and demonstrates that the phenomenon extends throughout the periodic table rather than being an isolated property of a single element.

3.8 Principal Findings

The full periodic-table audit establishes six principal observations:

1. Atomic radius systems exhibit systematic scaling relationships rather than random variation.
2. The Cordero (2008) Covalent Radius closely tracks the Empirical Atomic Radius across the periodic table.
3. Calculated Atomic Radius occupies a slightly larger but closely related radial scale.
4. van der Waals Radius forms a distinct outer radius family.
5. Radius relationships vary systematically by chemical family and period.
6. No single radius system provides complete coverage of all elements.

Collectively, these findings demonstrate that atomic radius variation is a genuine and structured empirical phenomenon. The principal radius systems do not represent independent or arbitrary measures of atomic size. Instead, they form reproducible hierarchical relationships across the periodic table. The existence of these ordered relationships motivates the central question addressed in the remainder of this work:

Why do multiple accepted radius definitions exist, and what physical regions of atomic structure do they represent?

4. Atomic Radius Variation as a Physical Observable

4.1 From Radius Definitions to Radius Structure

Section 2 established that multiple accepted radius definitions exist throughout atomic physics and chemistry. Each is derived using a different measurement methodology, experimental framework, or theoretical construct. Section 3 demonstrated that these radius systems are not randomly related. Instead, the principal radius definitions exhibit consistent scaling relationships across the periodic table. Empirical atomic radius and covalent radius occupy closely related radial scales, while van der Waals radius forms a distinct outer radius family. The existence of reproducible relationships between independent radius systems suggests that atomic radius variation is a structured phenomenon rather than a collection of unrelated measurement conventions. The natural next question therefore becomes:

What is the physical origin of atomic radius variation?

The conventional interpretation is that different radius definitions simply measure different operational properties of the atom. Under this view, radius variation arises primarily from methodology and definition rather than from a physical property of atomic structure itself.

The analysis presented here investigates an alternative possibility. Rather than treating the spread between radius systems as disagreement, the spread itself is treated as an observable quantity. For each element, the radius variation width was defined as:

$$\Delta R = R_{\max} - R_{\min}$$

where $R_{\max}$ and $R_{\min}$ represent the largest and smallest accepted radius values available for that element.

If radius variation is simply a consequence of measurement methodology, no systematic periodic-table structure would be expected within ΔR. Conversely, if the radius spread contains physical information, systematic organisation should emerge.

4.2 Radius Variation Exhibits Strong Periodic Structure

The results presented in Appendix E demonstrate that radius variation is not randomly distributed throughout the periodic table. The analysis reveals clear dependence upon both Group and Block classification. At Group level, mean radius variation ranges from approximately 29 pm in Group 12 to approximately 111 pm in Group 18. Such variation is substantial and far exceeds what would normally be expected from random measurement uncertainty. More significantly, the Block analysis reveals the existence of two statistically distinct populations.

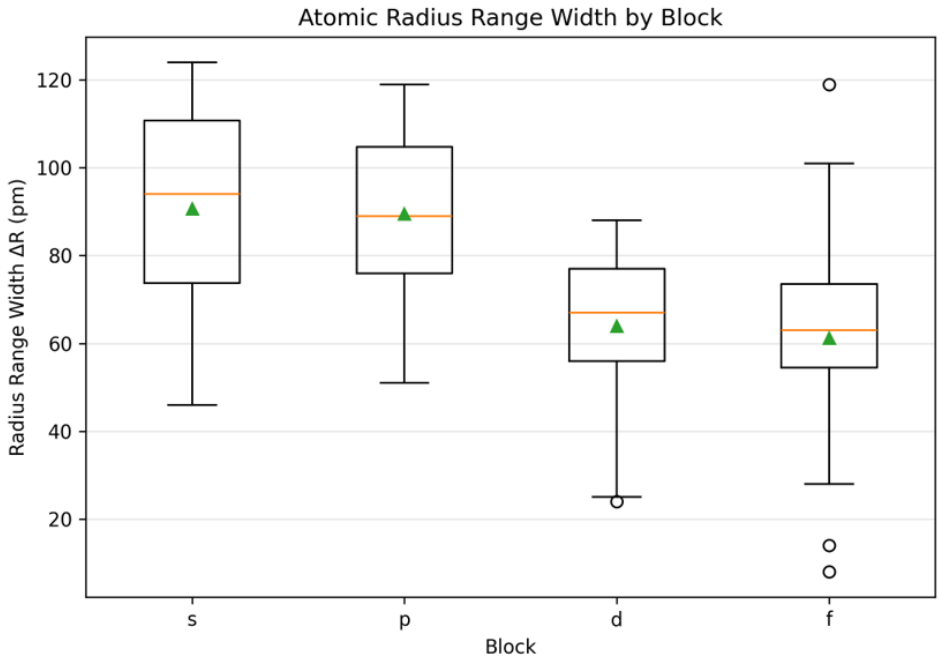


Figure 2. Atomic Radius Range Width by Block

Figure 2 shows a non-trivial statistically significant result; elements belonging to the s- and p-blocks exhibit mean radius variation widths of approximately:

$$\Delta R \approx 90\ pm$$

whereas elements belonging to the d- and f-blocks exhibit substantially smaller values:

$$\Delta R \approx 63\ pm$$

This separation is highly statistically significant. Both parametric and non-parametric hypothesis tests reject the null hypothesis that the four blocks originate from a common population. Pairwise comparisons further demonstrate that:

- s and p are statistically indistinguishable,
- d and f are statistically indistinguishable,
- s/p differs significantly from d/f.

The resulting effect size is large ($d = 1.29$), indicating that the separation is not merely statistically significant but physically substantial. The conclusion is unavoidable:

Atomic radius variation itself behaves as a structured periodic-table observable.

4.3 Radius Variation is Unlikely to be Measurement Error

The significance of this result extends beyond the numerical values alone. The analysis does not investigate any individual radius system in isolation. Instead, it examines the spread between independent radius systems derived using different methodologies. If the variation between accepted radius values arose solely from:

- measurement uncertainty,
- inconsistent experimental techniques,
- historical conventions,
- arbitrary definitional differences,

then the resulting radius spread would be expected to behave as noise, however;

1. Noise ***does not*** normally organise itself according to periodic-table architecture.
2. Noise ***does not*** naturally separate into statistically distinct s/p and d/f populations.
3. Noise ***does not*** produce highly significant Group and Block correlations.

Instead, the analysis demonstrates that radius variation contains reproducible periodic information. This strongly suggests that the various accepted radius definitions are not simply disagreeing about the size of the atom. Rather, they appear to be probing different regions of a common underlying atomic structure.

4.4 Evidence for a Finite Shell Thickness

The Carbon case study presented in Appendix B provides an important clue regarding the physical meaning of radius variation. Different radius definitions consistently occupy different radial positions. Triple-bond radii occupy the most internal positions. Single-bond and covalent radii occur at intermediate locations. Empirical and calculated radii cluster nearby. van der Waals radii consistently occupy the most external positions. The resulting pattern suggests that atomic shells may not be represented by infinitely thin boundaries.

Instead, the observational evidence is consistent with the existence of a finite radial shell region. Under this interpretation:

- smaller radius definitions probe more internal regions,
- larger radius definitions probe more external regions,
- the observed radius variation represents the thickness of the accessible shell domain.

The periodic dependence of ΔR implies that this shell thickness is itself a structured property of atomic architecture. Most importantly, the radius variation is not random. The shell thickness appears to vary systematically according to the underlying organisation of the periodic table.

4.5 Holland Atom Interpretation

Within the Holland Atom framework, the observed behaviour acquires a natural physical interpretation. The Holland Atom does not describe atomic shells as infinitesimally thin surfaces. Instead, shells are represented as resonance-locked standing-wave structures possessing finite radial extent [30,31].

A standing wave is characterised not only by a mean radius but also by an amplitude. The shell therefore occupies a band of radial space rather than a mathematically sharp boundary. Under this interpretation, the various accepted radius definitions may be sampling different locations within the standing-wave profile:

- inner boundary regions,
- mean-density regions,
- outer boundary regions,
- interaction regions associated with neighbouring atoms.

The quantity $\Delta R = R_{\max} - R_{\min}$ therefore becomes physically meaningful. Rather than representing disagreement between measurement systems, it may represent an observable manifestation of **shell-band thickness**. The statistically significant dependence of ΔR upon Group and Block classification is consistent with the possibility that standing-wave amplitude is not constant but varies according to shell structure, resonance geometry, and periodic-table architecture. This interpretation remains a hypothesis. However, it provides a direct causal explanation for both the existence of multiple accepted radius definitions and the structured periodic behaviour observed throughout the dataset.

4.6 Conclusions

The combined evidence from Sections 2–4 supports a significant revision in how atomic radius variation may be interpreted. The existence of multiple radius definitions is well established. What has not previously been recognised is that the spread between those definitions behaves as a structured and statistically significant periodic-table observable.

The radius variation width $\Delta R = R_{\max} - R_{\min}$ exhibits strong dependence upon Group and Block classification and separates into two statistically distinct populations corresponding broadly to the s/p and d/f regions of the periodic table. These observations indicate that atomic radius variation contains genuine structural information and is unlikely to be explained solely as measurement disagreement or methodological uncertainty. The results therefore support the hypothesis that accepted **radius definitions are probing different regions of a physically extended shell structure**.

Within the Holland Atom framework, this naturally suggests that the observed radius variation may represent the measurable radial thickness of resonance-locked standing-wave shells. If correct, the accepted radius systems are not competing descriptions of atomic size but complementary measurements sampling different regions of the same physical structure.

5. The Holland Atom Interpretation of Atomic Radius Variation

5.1 Introduction

Sections 2–4 established three principal findings.

1. Multiple accepted atomic radius definitions exist throughout atomic physics and chemistry, each derived from different experimental methodologies, theoretical frameworks, and physical interpretations.
2. These radius systems exhibit systematic and reproducible scaling relationships across the periodic table rather than random variation.
3. The spread between accepted radius definitions $\Delta R = R_{\max} - R_{\min}$ behaves as a statistically significant periodic-table observable, exhibiting strong dependence upon Group and Block structure.

The evidence presented thus far is entirely empirical and does not depend upon any particular atomic model. The question now becomes:

What physical mechanism could give rise to the observed hierarchy of radius measurements and the structured variation between them?

This section introduces the Holland Atom interpretation and proposes a causal explanation for the observed behaviour.

5.2 The Holland Atom Research Programme

The present work forms part of the wider Holland Atom research programme, itself derived from the broader General Expanse Tension Theory (GETT) framework [24-29]. GETT proposes that matter, gravity, and physical structure emerge from interactions within the Holland Field (Φ), an all-pervading density-dependent field whose behaviour varies across multiple physical regimes [21-23]. Within this framework, the Holland Atom was developed as a reconstruction of atomic structure from first principles using resonance-lock dynamics rather than the conventional Copenhagen interpretation. The first Holland Atom paper demonstrated that a resonance-lock shell architecture can reproduce a wide range of observed atomic behaviours [31], including:

- shell ordering,
- shell capacities,
- shell accessibility,
- shell filling sequences,
- shell contraction,
- anomalous electronic configurations,
- periodic-table structure.

Importantly, the Holland Atom does not treat shells as collections of orbiting particles occupying probabilistic orbitals. Instead, shells emerge as resonance-locked standing-wave structures within the Holland Field. This distinction is fundamental to the interpretation developed in the present paper.

5.3 Shells as Resonance-Locked Standing Modes

The Holland Atom proposes that atomic structure consists of nested resonance-locked shell modes. Each shell corresponds to a standing mode formed through resonance within the local field environment surrounding the nucleus. A standing mode is not an infinitesimally thin surface. Instead, every standing wave possesses:

- a mean position,
- a finite amplitude,
- an inner boundary,
- an outer boundary.

Consequently, the shell occupies a finite radial domain rather than a single geometric radius. This concept is illustrated schematically in Figure 3.

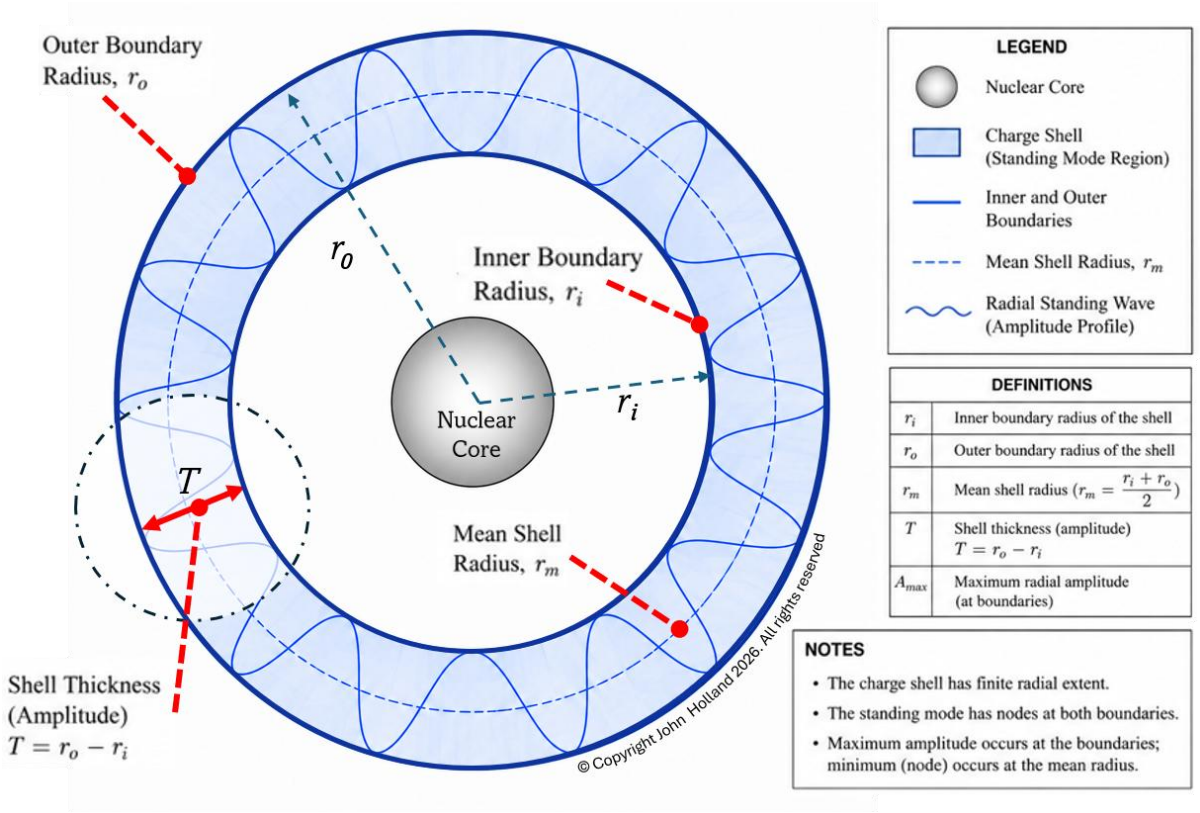


Figure 3. Holland Atom shell-band geometry and standing-mode amplitude.

Figure 3 shows a schematic representation of a single Holland Atom charge shell. The shell is modelled as a finite radial envelope bounded by an inner radius (r_i) and outer radius (r_o), giving a shell thickness $T = r_o - r_i$. The standing mode oscillates symmetrically between the two boundaries, with the mean shell radius (r_m) defining the midline of the shell-band. Under the Holland Atom interpretation, atomic charge shells are resonance-locked standing modes whose finite radial extent provides a physical basis for the observed variation between accepted atomic radius definitions. Different radius systems may therefore sample different regions within the same shell envelope rather than representing conflicting measurements of atomic size.

Therefore there are really **three related but distinct quantities**:

1. **Radius range**: $\Delta R = R_{max} - R_{min}$

2. **Half-range amplitude:** $A = \frac{\Delta R}{2}$

3. **Shell thickness**: $T = 2A = \Delta R$

and these have slightly different physical meanings. For statistical convenience the present study uses the half-range amplitude, as the primary amplitude metric. However, within the Holland Atom interpretation the physically meaningful quantity is the full shell thickness,

$$T = r_o - r_i = 2A = \Delta R$$

representing the total radial extent of the resonance-locked shell. Throughout the remainder of this paper, measurements of A may therefore be interpreted directly as measurements of shell thickness through the relationship $T = 2A$.

Within such a structure, there is no requirement for a unique atomic radius. Different experimental methodologies may naturally sample different regions of the standing-mode profile.

5.4 Reinterpreting Atomic Radius Measurements

The conventional interpretation assumes that the various radius systems represent different operational definitions of atomic size. The Holland Atom proposes a different interpretation. The various radius measurements may all be correct. They may simply be measuring different regions of the same shell structure.

The shell-region assignments proposed in Table 4 should be interpreted as a working physical model rather than a uniquely determined result of the radius audit. The ordering of the radius families is directly supported by the observational analyses presented in Sections 2–4 and Appendices B–D. However, the assignment of specific radius families to particular shell regions represents an interpretation of those observations within the Holland Atom framework. Alternative physical interpretations of the same ordering may be possible. The significance of Table 4 is therefore not that the shell locations are proven, but that the observed radius hierarchy is consistent with a finite shell-band structure containing distinct radial regimes.

Under this interpretation:

Radius Type	Potential Shell Region Sampled
Triple-bond covalent radius	Inner shell region
Double-bond covalent radius	Inner-intermediate region
Single-bond covalent radius	Intermediate region
Empirical atomic radius	Intermediate shell region
Calculated atomic radius	Mean-density region
van der Waals radius	Outer shell boundary

Table 4. Proposed Shell Regions Sampled (Holland Atom Interpretation)

Table 4 proposes the Holland Atom interpretation for mapping of radius type to shell region, though the exact mapping remains a subject for future investigation. However, the Carbon analysis presented in Appendix B demonstrated a consistent ordering of radius values that is compatible with such an interpretation.

The key point is that the radius hierarchy appears structured rather than arbitrary.

5.5 Radius Variation as Shell-Band Width

Sections 4 and Appendix E demonstrated that radius variation behaves as a structured periodic-table observable. Appendix F extended this analysis by decomposing the observed radius variation into:

$$R_{mid} = \frac{R_{\max} + R_{\min}}{2}$$

and:

$$A = \frac{R_{\max} - R_{\min}}{2}$$

where A represents the half-range amplitude of the observed radius band. The analysis revealed several remarkable observations. The shell-band half-range amplitude, $A = \Delta R/2$, provides a measure of the radial extent of the observed radius envelope. The global mean amplitude is:

$$A \approx 38 \text{ pm}$$

The amplitude exhibits strong periodic-table dependence. Most significantly, the periodic-table blocks separate into two statistically distinct amplitude populations:

$$A_{s+p} \approx 45 \text{ pm}$$

$$A_{d+f} \approx 31 \text{ pm}$$

with:

$$p = 8.6 \times 10^{-9}$$

and a large effect size:

$$d = 1.29$$

These observations indicate that the radius variation is not simply a consequence of measurement uncertainty. Instead, the amplitude itself behaves as a structured physical quantity.

5.6 Standing-Mode Amplitude Hypothesis

The central hypothesis emerging from this work is therefore straightforward, and is visualised in Figure 4.

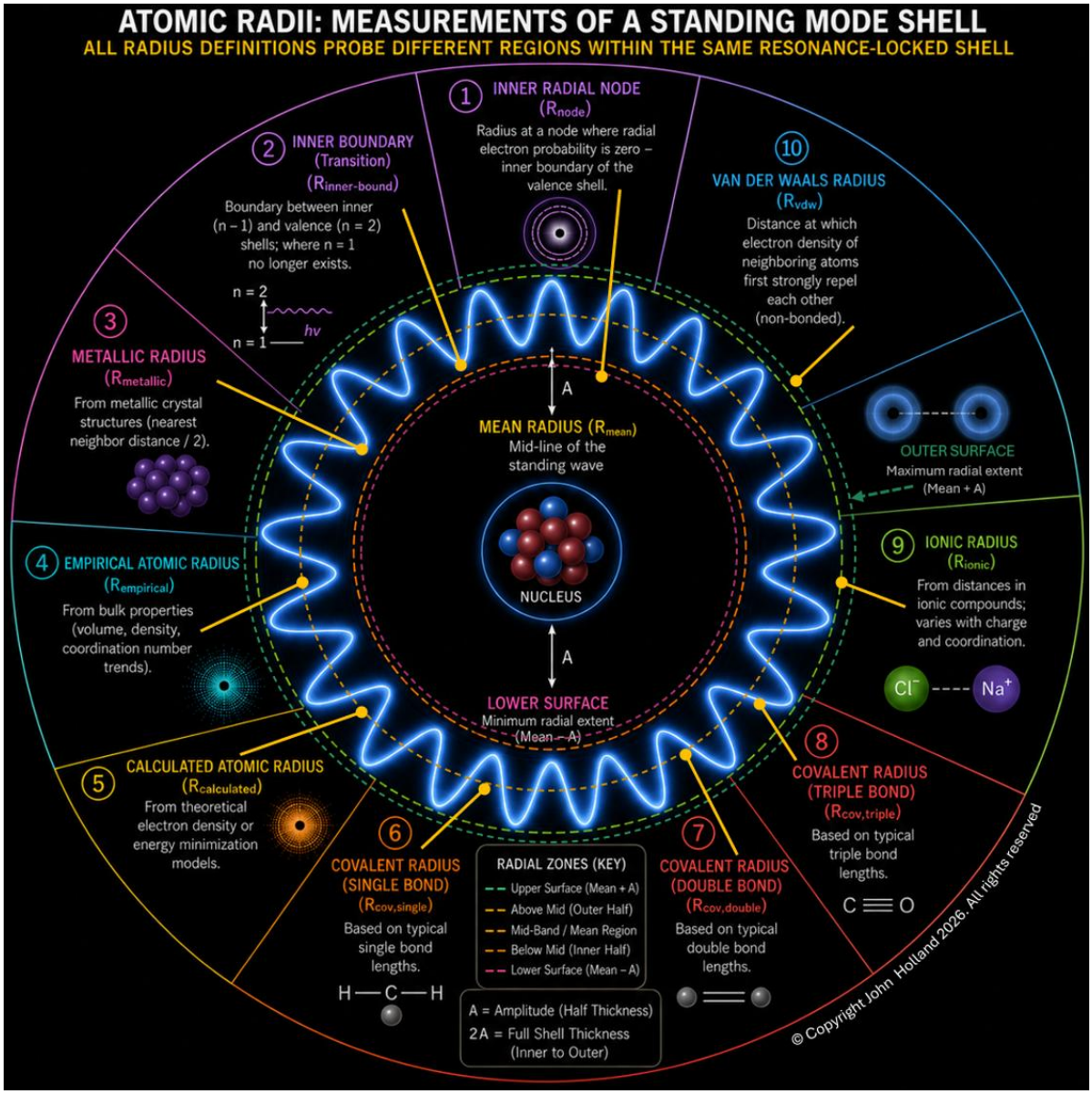


Figure 4. Unified Interpretation of Atomic Radius Measurements within a Resonance-Locked Shell

Figure 4 *illustrates the central hypothesis proposed in this work: the various accepted atomic radius definitions do not represent conflicting measurements of atomic size but rather probe different radial regions within the same finite-thickness shell structure. Appendix G demonstrates that the outer interaction envelope, represented primarily by van der Waals radii, behaves differently from the internal radius family. Consequently, the shell-band illustrated here should be interpreted as a structured radial envelope containing both internal and outer-boundary observational regimes rather than a single uniform shell thickness. The shell envelope appears to contain multiple radial regimes. Radius systems associated with bonding and crystal structure define an internal radius family, while van der Waals radii define an outer interaction envelope. The standing-mode interpretation proposed here therefore refers not to a single homogeneous shell thickness, but to a structured radial envelope containing multiple observational layers.*

The shell is represented as a resonance-locked standing mode possessing a mean radius (R_{mean}), a lower surface ($R_{mean} - A$), and an outer surface ($R_{mean} + A$), where A is the standing-mode amplitude. Within the Holland Atom framework, the shell-band half-range amplitude is interpreted as the standing-mode amplitude of the resonance-locked shell. The full shell thickness is therefore:

$$2A = R_{outer} - R_{inner}$$

Unlike conventional depictions that treat atomic shells as infinitesimally thin boundaries, the standing-mode interpretation proposes that each shell occupies a finite radial domain bounded by physically meaningful inner and outer limits. The radius definitions examined throughout this paper are positioned according to the Carbon audit (Appendix B) and the radius variation analyses (Appendices D–F). Each radius type samples a different region of the shell band:

- **Inner boundary indicators** include the inner radial node and shell-transition boundary.
- **Lower-mid shell measurements** include metallic radius and calculated atomic radius.
- **Mid-band measurements** include empirical atomic radius and single-bond covalent radius.
- **Upper-mid shell measurements** include double-bond and triple-bond covalent radii.
- **Outer boundary measurements** include ionic radius and van der Waals radius.

The placement of these measurements reflects the observed ordering of accepted radius values rather than arbitrary assignment. Figure 4 provides a unified interpretation of the long-standing multiplicity of atomic radius definitions. Under this hypothesis, the spread between accepted radius values is not measurement disagreement but a consequence of different experimental, observational, and theoretical methods sampling different locations within a physically extended shell structure. This interpretation is supported by the statistical analyses presented in Appendices E and F, which demonstrate that the radius range width is strongly correlated with periodic-table architecture and separates into statistically distinct block populations. The observed radius variation therefore behaves as a structured physical observable rather than random measurement uncertainty. The observed radius variation may represent the measurable radial amplitude of resonance-locked atomic shells. Under this interpretation:

$$\Delta R = R_{\max} - R_{\min}$$

corresponds to the observable shell-band thickness. The various accepted radius definitions are therefore not competing descriptions of atomic size. Instead, they represent measurements, calculations, deductions, and inferences sampling different locations within a physically extended shell structure. The significance of this interpretation is substantial. For over a century, atomic radius variation has generally been regarded as an unavoidable consequence of differing measurement methodologies. The present results suggest an alternative possibility: **the variation itself may be the signal.**

5.7 Relationship to the 1920s Pre-Copenhagen Model

One of the most striking aspects of this interpretation is its historical consistency. Prior to the establishment of the Copenhagen interpretation, many early atomic models envisioned atomic structure in terms of physical wave phenomena, resonances, and standing modes.

Standing-wave explanations of atomic quantisation have deep roots in the development of quantum theory. De Broglie and Schrödinger both employed wave-based descriptions in which allowed atomic states emerged from resonant conditions and standing-wave solutions. However, these approaches did not identify a physical field responsible for generating or supporting the standing modes, nor did they provide a density-dependent mechanism capable of determining shell radii and shell capacities [1-6].

Within General Expanse Tension Theory (GETT), the Holland Field Φ is proposed as the all-pervading physical medium supporting these resonance-locked standing modes, while local density-dependent Holland–Higgs coupling provides the mechanism by which shell geometry, shell spacing, and atomic dimensions emerge. This hypothesis was first introduced in GETT framework's application to the 1920's pre-Copenhagen empirical dataset, seeking a scalar field – "*realism*" - alternative to the probability-based interpretation which followed [30].

The Holland Atom revisits this earlier conceptual direction using modern empirical evidence and contemporary statistical analysis. The present work demonstrates that the observed spread between accepted radius definitions contains structured periodic information. This behaviour is precisely what would be expected if atomic shells possess finite radial extent and standing-mode amplitude. Thus, the empirical evidence obtained in Sections 2–4 and Appendices B–F is consistent with a modern resonance-lock interpretation of atomic structure.

5.8 Conclusions

The analyses presented throughout this paper lead to a simple but potentially profound conclusion. The existence of multiple accepted atomic radius definitions may not represent disagreement regarding the size of the atom. Instead, the various radius systems may be observing different regions of a physically extended shell structure. The periodic dependence of radius variation, the existence of statistically distinct amplitude populations, and the characteristic shell-band widths observed throughout the periodic table are all consistent with this interpretation. Within the Holland Atom framework, atomic shells are resonance-locked standing modes possessing finite radial extent.

Under this model:

- atomic radius variation becomes physically meaningful,
- radius spread becomes an observable quantity,
- shell-band amplitude becomes measurable,
- and the long-standing multiplicity of atomic radius definitions acquires a common causal explanation.

The hypothesis advanced here is therefore that accepted atomic radius measurements are not conflicting descriptions of atomic size. Rather, they are complementary observations of different locations within a finite-amplitude resonance-locked shell structure.

6. Determination of the Holland Atom Radius Standard

6.1 Introduction

Sections 2–5 established that atomic radius is not represented by a single universally accepted quantity. Instead, multiple radius definitions coexist throughout atomic physics and chemistry, each derived using different experimental methodologies, theoretical frameworks, and physical assumptions. The analyses presented in this paper demonstrate that these radius definitions are not arbitrary. They exhibit systematic relationships, reproducible scaling behaviour, and statistically significant periodic structure. Furthermore, the observed spread between radius definitions appears consistent with the interpretation that different measurement methodologies probe different radial regions of a finite shell structure. While this conclusion provides an explanation for the long-standing multiplicity of atomic radius definitions, it introduces a practical challenge for the Holland Atom research programme.

Future Holland Atom papers will require a single standard radius metric for:

- shell geometry analysis,
- charge-density calculations,
- isotopic scaling studies,
- periodic behaviour investigations,
- bonding analysis,
- compound geometry prediction,
- calibration and validation exercises.

A standard reference radius must therefore be selected. The purpose of this section is to determine which radius definition should serve as the primary geometric reference throughout the Holland Atom research portfolio.

6.2 Requirements for a Reference Radius

The objective is not to determine which radius definition is most useful for a particular branch of chemistry or physics. Rather, the objective is to identify which radius best represents the characteristic geometry of the shell itself. A suitable Holland Atom reference radius should satisfy the following criteria:

1. Broad periodic-table coverage.
2. Strong experimental foundation.
3. Minimal dependence upon specific bonding environments.
4. Minimal dependence upon crystal structure.
5. Minimal dependence upon theoretical assumptions.
6. Stable positioning within the shell band.
7. Consistent periodic behaviour.
8. Physical relevance to intrinsic atomic geometry.

Radius systems that primarily describe interaction limits, crystal packing, or highly environment-dependent effects are less suitable as universal geometric references.

6.3 Assessment of Candidate Radius Systems

Empirical Atomic Radius

The empirical atomic radius is derived from observed atomic dimensions and periodic trends [9,10].

Advantages include:

- Direct experimental foundation.
- Long historical usage.
- Strong periodic behaviour.
- Representation of the neutral atom.
- Minimal dependence upon specific bonding environments.

Disadvantages include:

- Missing values for a limited number of elements.
- Dependence upon interpretation of experimental datasets.

However, throughout the analyses presented in this paper, empirical atomic radius consistently occupies a central position within the shell band.

Calculated Atomic Radius

Calculated atomic radius is obtained from theoretical electron-density models [9,10].

Advantages include:

- Broad coverage.
- Consistent methodology.
- Useful theoretical interpretation.

Disadvantages include:

- Model dependence.
- Dependence upon assumptions within quantum calculations.
- Not directly measured.

The calculated radius appears to occupy a similar region to empirical atomic radius but lacks the advantage of direct observational grounding.

Covalent Radius (2008)

The Cordero et al. (2008) covalent radius dataset represents one of the most comprehensive modern radius compilations [12].

Advantages include:

- Extensive coverage.
- High-quality modern dataset.
- Strong periodic behaviour.
- Broad acceptance.

Disadvantages include:

- Derived from bonding environments.
- Represents interaction geometry rather than intrinsic shell geometry.

The analysis presented in Sections 2–4 demonstrates that covalent radii occupy a consistent region within the shell band but are inherently linked to bonding behaviour.

Metallic Radius

Metallic radius is derived from metallic crystal structures [18].

Advantages include:

- Experimentally measurable.
- Useful for metallic systems.

Disadvantages include:

- Not applicable to all elements.
- Strong dependence upon crystal packing.

Its limited applicability prevents adoption as a universal reference radius.

van der Waals Radius

van der Waals radius represents the non-bonded interaction limit between neighbouring atoms [8-16].

Advantages include:

- Physically meaningful.
- Extensive dataset.
- Strong experimental foundation.

Disadvantages include:

- Represents the outer interaction boundary of the shell.
- Not representative of average shell geometry.

The analyses throughout this paper consistently place van der Waals radius near the outermost accessible region of the shell band.

It therefore functions naturally as a shell-boundary indicator rather than a shell-centre indicator.

6.4 Position Within the Shell Band

The Carbon case study presented in Appendix B provides a useful illustration. Across the available radius definitions, empirical atomic radius consistently occupies a central position within the observed shell band. It is neither:

- an inner-boundary metric,
- nor an outer interaction-boundary metric.

Instead, it lies close to the midpoint of the shell region sampled by the various radius systems. This observation is important because the purpose of the Holland Atom reference radius is to represent the characteristic shell geometry rather than a particular interaction mechanism. A radius definition located near the centre of the shell band is therefore preferable to one located near either boundary.

6.5 Selection of the Holland Atom Radius Standard

Based upon the evidence presented throughout this paper, the empirical atomic radius is selected as the standard geometric radius for the Holland Atom framework. The Holland Atom reference radius is therefore defined as:

$$R_{HA} = R_{empirical}$$

where:

$$R_{HA}$$

represents the characteristic geometric radius of the shell. This choice is justified because empirical atomic radius:

- possesses strong experimental grounding,
- exhibits stable periodic behaviour,
- occupies a central position within the shell band,

- is largely independent of specific chemical environments,
- and most closely represents the characteristic geometry of the neutral atom.

6.6 Treatment of Missing Values

A small number of elements lack empirical atomic radius values within the available datasets. Examples include helium and xenon, where conventional empirical atomic radius values are not reported. For such elements, the Holland Atom programme will adopt the following hierarchy:

1. Published empirical atomic radius where available.
2. Interpolated value derived from validated periodic trends.
3. Value inferred from calibrated relationships between empirical and covalent radius datasets.

The selected replacement values must remain explicitly identified as inferred rather than directly measured.

6.7 Implications for Future Holland Atom Papers

The adoption of a single reference radius provides a consistent geometric foundation for the remainder of the Holland Atom research programme. Future papers will therefore use:

$$R_{HA} = R_{empirical}$$

as the standard shell radius when calculating:

- shell scaling laws,
- charge-density relationships,
- isotopic geometry shifts,
- periodic-table behaviour,
- bonding geometry,
- compound structure,
- and resonance-lock predictions.

Importantly, the adoption of a reference radius does not imply that alternative radius definitions are incorrect. The principal conclusion of this paper is precisely the opposite. The various accepted radius definitions are interpreted as valid measurements of different regions within a finite shell band. The empirical atomic radius is selected only because it provides the most suitable geometric reference point within that band.

6.8 Conclusions

The analyses presented throughout this paper demonstrate that atomic radius is not adequately represented by a single universally correct measurement. Instead, multiple accepted radius definitions probe different regions of a physically extended shell structure. Nevertheless, a standard geometric reference is required for future Holland Atom development. After assessment of the major radius systems, empirical atomic radius is selected as the Holland Atom reference radius.

This radius occupies a stable central position within the shell band, possesses strong observational grounding, and provides the most appropriate representation of characteristic shell geometry.

The selection of Empirical Atomic Radius as the Holland Atom reference radius should not be interpreted as implying that alternative radius systems are incorrect or physically inferior. On the contrary, the analyses presented throughout this work demonstrate that the major radius families

exhibit systematic and reproducible relationships. Consequently, once a Holland Atom geometric model has been established using the reference radius R_a, the corresponding values for other accepted radius systems may be recovered through the empirically determined scaling relationships identified in this study. The choice of reference radius therefore establishes a common geometric baseline rather than excluding the continued use of alternative radius definitions. In this sense, the selection of R_a represents a coordinate choice within a broader radius-family framework rather than the selection of a uniquely "correct" atomic radius.

The remainder of the Holland Atom research programme will therefore adopt empirical atomic radius as the standard geometric radius while retaining the broader shell-band interpretation developed throughout this paper.

7. Empirical Characterisation of Atomic Shell-Band Structure

7.1 Introduction

The analyses presented throughout this paper have examined the long-standing problem of atomic radius variation. Traditionally, the existence of multiple accepted atomic radius definitions has been interpreted as a consequence of differing measurement methodologies, differing physical assumptions, and differing operational definitions of atomic size. The results obtained here support a different interpretation. The evidence demonstrates that atomic radius variation is not random, is not uniformly distributed throughout the periodic table, and contains statistically significant periodic structure. Furthermore, the various radius definitions exhibit reproducible relationships and occupy consistent positions within a finite shell-band framework.

The objective of this section is to consolidate the empirical findings of the study and quantify the observable properties of the shell band itself.

7.2 Summary of Experimental Findings

The principal empirical findings established throughout the paper may be summarised as follows.

Finding 1 — Multiple Radius Definitions Form an Ordered Hierarchy

The radius audit demonstrated that accepted atomic radius definitions occupy reproducible positions relative to one another.

The ordering observed for Carbon, and subsequently supported by broader periodic-table analysis, reveals a structured radial hierarchy extending from inner-boundary indicators through intermediate shell measures to outer interaction limits. This hierarchy is not random and appears consistent across multiple elements.

Finding 2 — Radius Definitions Exhibit Systematic Scaling Relationships

Normalised radius analysis demonstrated that the major radius systems exhibit reproducible scaling behaviour relative to empirical atomic radius. The radius definitions therefore behave as related observables rather than independent measurements.

Finding 3 — Radius Variation Contains Periodic Information

The radius spread, $\Delta R = R_{\max} - R_{\min}$, was shown to vary systematically throughout the periodic table. Both Group and Block analyses revealed statistically significant organisation. The spread between radius definitions therefore contains genuine periodic information.

Finding 4 — Radius Variation Forms Distinct Block Populations

The periodic-table blocks separate into two statistically distinct populations. Elements belonging to the s- and p-blocks exhibit substantially larger radius variation than d- and f-block elements. The effect is highly significant and cannot reasonably be attributed to measurement uncertainty.

Finding 5 — Radius Variation Behaves as an Independent Observable

Appendix F demonstrated that shell-band amplitude is effectively uncorrelated with shell-band midpoint. The width of the shell band therefore appears to behave as an independent structural quantity rather than a simple consequence of atomic size.

7.3 Measured Shell-Band Width

The full periodic-table audit yielded a mean shell-band width ($\Delta R = 2A = T$) of:

$$\bar{\Delta R} = 75.2 \text{ pm}$$

with a median value of:

$$74.5 \text{ pm}$$

and a standard deviation of:

$$24.9 \text{ pm}$$

This quantity represents the observed radial spread between accepted radius definitions. Under the shell-band interpretation developed in this paper, it corresponds to the measurable thickness of the shell region. The existence of a characteristic shell-band width is itself significant. If radius variation were purely methodological, no characteristic scale would necessarily be expected. Instead, the analysis reveals a reproducible width scale throughout the periodic table.

7.4 Measured Standing-Mode Amplitude

The shell-band amplitude was defined as:

$$A = \frac{\Delta R}{2}$$

where A represents the half-range amplitude of the shell band. The analysis yielded:

$$\bar{A} = 37.6 \text{ pm}$$

with a median value of:

$$37.3 \text{ pm}$$

This value represents the average radial displacement between the shell midpoint and the shell boundary. Within the Holland Atom interpretation, this quantity may be interpreted as the observable

amplitude of the resonance-locked standing mode. Importantly, this value emerges directly from the observational datasets and is not introduced as a fitted parameter.

7.5 Shell-Band Amplitude is Independent of Atomic Size

A natural explanation for the observed radius variation is that larger atoms simply possess proportionally larger radius spreads. Under such a model, shell-band amplitude would be expected to increase with atomic size. This hypothesis was tested by comparing shell-band amplitude,

$$A = \frac{\Delta R}{2}$$

against shell-band midpoint,

$$R_{mid} = \frac{R_{max} + R_{min}}{2}$$

using the complete seven-radius-family dataset. The resulting correlation was:

$$r(A, R_{mid}) = -0.028$$

with:

$$p = 0.785$$

indicating no statistically significant relationship. The full observable radius envelope therefore exhibits an amplitude that is effectively independent of atomic size. Importantly, this result applies specifically to the complete radius envelope incorporating empirical, calculated, covalent, metallic, and van der Waals radius systems. As shown in Appendix G, removal of the van der Waals radius family restores a significant positive correlation between amplitude and midpoint, indicating that the observed independence is a property of the full observable shell envelope rather than the internal radius family alone. The observed shell-band amplitude is therefore effectively independent of atomic size. This result is important because it eliminates the most obvious geometric explanation for the observed radius variation. The shell-band width does not arise simply because larger atoms possess larger radii. Instead, shell-band amplitude behaves as an independent atomic property. The absence of correlation within the full radius envelope remains significant because it demonstrates that the total observable shell-band width is not a simple geometric consequence of increasing atomic size.

7.6 Block Dependence of Shell Amplitude

The block dependence of shell-band amplitude depends upon which radial regime is considered. Using the complete seven-radius-family dataset, including van der Waals radii, s- and p-block elements exhibit the largest observable shell-envelope amplitudes. However, Appendix G demonstrates that removal of the van der Waals radius family reverses this relationship, with d- and f-block elements exhibiting the larger internal radius-family spread. The block dependence therefore reflects a structured shell envelope containing distinct internal and outer-boundary components rather than a single amplitude parameter. The measured mean amplitudes are:

Block	Mean Amplitude A (pm)
s	45.3
p	44.7
d	32.0
f	30.6

Table 5. Mean Amplitude A by Block

The data therefore separate naturally into two populations.

Wide-Amplitude Population

$$A_{s+p} = 44.9 \text{ pm}$$

Narrow-Amplitude Population

$$A_{d+f} = 31.4 \text{ pm}$$

The difference between these populations is highly statistically significant:

$$p = 8.6 \times 10^{-9}$$

with a large effect size:

$$d = 1.29$$

The shell-band amplitude therefore varies systematically according to atomic architecture.

7.7 Group Dependence of Shell Amplitude

Group analysis further demonstrates that shell-band width is not uniformly distributed throughout the periodic table. The largest mean shell-band widths occur in:

- Group 18 (Noble Gases)
- Group 17 (Halogens)
- Group 16 (Chalcogens)
- Group 14 (Carbon Group)

The smallest mean shell-band widths occur within the transition-metal region. The amplitude therefore appears to contain information regarding atomic family structure. This observation reinforces the conclusion that shell-band width is a genuine atomic property rather than a consequence of measurement methodology.

7.8 Shell Dependence

The present study has established dependence upon Group and Block classification. However, the available data also suggest the possibility of a deeper relationship with shell architecture itself.

The Holland Atom framework naturally predicts that shell-band properties may vary with n, where n represents the highest occupied shell. The present work has not attempted to derive a shell-number scaling law for amplitude. Nevertheless, the identification of a measurable shell-band width now makes such investigations possible. Future studies will therefore examine:

$$A(n)$$

as a potential shell-scaling observable.

7.9 Radius Variation as a Physical Observable

The most important outcome of this work is not the precise numerical value of the amplitude. Rather, it is the discovery that the radius variation itself behaves as a measurable physical quantity.

Historically, the spread between accepted radius definitions has generally been regarded as a limitation of measurement methodology. The present analysis demonstrates that the spread exhibits:

- periodic structure,
- family dependence,
- block dependence,
- statistical significance,
- characteristic amplitude scales.

These properties are more consistent with a physical observable than with random measurement disagreement. Radius variation therefore appears to contain genuine structural information about atomic architecture.

7.10 Holland Atom Interpretation

Within the Holland Atom framework, the results presented here acquire a natural interpretation. The shell is not viewed as an infinitesimally thin surface. Instead, the shell occupies a finite radial domain bounded by lower and upper surfaces and characterised by a measurable amplitude. The various accepted radius definitions become measurements of different regions within that domain.

The quantity $\Delta R = R_{\max} - R_{\min}$ represents the observable shell thickness. Under this interpretation, the multiple accepted radius definitions are not conflicting descriptions of atomic size but complementary observations of different regions within the same resonance-locked shell structure.

7.11 Conclusions

The analyses presented throughout this paper support four principal conclusions.

First, atomic radius variation contains statistically significant periodic structure.

Second, the spread between accepted radius definitions behaves as a measurable atomic observable.

Third, the shell-band possesses a characteristic thickness of approximately:

$$75 \text{ pm}$$

corresponding to a characteristic half-range amplitude of approximately:

$$38 \text{ pm}$$

Fourth, shell-band amplitude varies systematically according to periodic-table architecture, with s- and p-block elements exhibiting substantially larger amplitudes than d- and f-block elements.

Taken together, these observations suggest that accepted atomic radius measurements are sampling different regions of a physically extended shell structure. Within the Holland Atom framework, this structure is interpreted as a resonance-locked standing mode whose finite amplitude gives rise to the observed hierarchy of atomic radius measurements.

8. Discussion: Interpretation and Significance of the Radius-Variation Results

8.1 Purpose of the Discussion

The purpose of this paper was to test whether the long-established variation between accepted atomic radius definitions could contain physical information about atomic shell structure.

The starting hypothesis was derived from the Holland Atom framework introduced in Holland Atom Paper 1: that atomic charge shells are not infinitesimally thin surfaces, but resonance-locked standing-mode structures triggered by nuclear mass loading. Under that interpretation, a shell should possess finite radial depth. The spread between accepted radius definitions may therefore provide evidence of shell thickness, shell amplitude, or shell envelope structure. The analyses presented in Sections 2–7 and Appendices A–G now allow this hypothesis to be assessed against empirical and calculated radius datasets.

8.2 What Is Already Known

It is already well established that atomic radius is not a uniquely defined quantity. Different radius systems are used for different purposes, including empirical atomic radius, calculated atomic radius, covalent radius, metallic radius, ionic radius, and van der Waals radius. It is also well known that covalent radii, van der Waals radii, and calculated radii arise from different physical and methodological contexts. Cordero et al. explicitly derived a revised covalent-radius dataset from crystallographic data for most elements up to $Z = 96$, and noted that the radii show periodic dependence, including transition-metal and lanthanide contraction effects.

van der Waals radii are also known to represent a distinct class of radius associated with non-bonded interaction behaviour rather than covalent bonding geometry. Modern work relates van der Waals radius to atomic polarizability and dispersion interactions, reinforcing that R_{vdW} is not merely a larger version of covalent radius but reflects a different physical interaction regime.

Therefore, the fact that radius definitions differ is not new. What is new in this paper is the treatment of the **spread between radius definitions** as a measurable quantity in its own right.

8.3 What This Paper Adds

The central contribution of this paper is not the observation that atomic radii vary. The central contribution is the finding that atomic radius variation is structured. Rather than treating the differences between accepted radius values as methodological inconvenience, the paper defines:

$$\Delta R = R_{\max} - R_{\min}$$

and:

$$A = \frac{\Delta R}{2}$$

and asks whether these quantities behave randomly or systematically. The answer is clear. They behave systematically.

The full periodic-table audit found:

$$\bar{\Delta R} = 75.2 \text{ pm}$$

and:

$$\bar{A} = 37.6 \text{ pm}$$

across the analysed element set. This suggests that the spread between accepted radius definitions has a characteristic physical scale.

8.4 The Most Important Empirical Result

The most striking result is not simply that A exists. The most striking result is that A is essentially independent of radius-band midpoint:

$$r(A, R_{mid}) = -0.028 \text{ with } p = 0.785$$

This is a critical finding. It rules out the simplest objection: that larger atoms merely possess larger radius spreads because they are larger. If the radius range were simply a proportional scaling effect, then Ashould increase with R_{mid}. It does not. The shell-band half-range is therefore not merely a trivial consequence of atomic size. This result strongly supports the interpretation that radius variation behaves as an independent structural observable.

8.5 Group and Block Dependence

The radius-range analysis further demonstrated that ΔR and A are strongly dependent upon periodic-table architecture. The most important separation occurs by block. With all radius families included, the mean half-range amplitudes are shown in Table 5.

Thus, s- and p-block elements form a wider-amplitude population, while d- and f-block elements form a narrower-amplitude population. The combined comparison gives:

$$A_{s+p} = 44.9 \text{ pm}$$
$$A_{d+f} = 31.4 \text{ pm}$$

with:

$$p = 8.6 \times 10^{-9}$$

and effect size:

$$d = 1.29$$

This is not a marginal effect. It is a large and statistically significant separation. The conclusion is that radius variation is not merely element-specific. It is structured by periodic-table family.

8.6 Meaning of the van der Waals Robustness Test

The most important limitation identified during review was that the full radius-band width is often dominated by R_{vdW}, because van der Waals radius is frequently the largest available radius value.

Appendix G addressed this directly. When van der Waals radius was removed, the mean radius range fell from:

75.2 pm

to:

48.2 pm

and the mean half-range amplitude fell from:

37.6 pm

to:

$$24.1 \text{ pm}$$

This confirms that R_{vdW} is a major contributor to the full observable shell envelope. However, the signal did not disappear. Block dependence remained highly significant:

$$p = 8.74 \times 10^{-12}$$

for the no-vdW ANOVA. The interpretation therefore becomes more refined. The full radius-band structure contains at least two components:

1. An outer interaction-envelope component, dominated by van der Waals radius.
2. An internal radius-family component, dominated by empirical, calculated, covalent, metallic, and bond-order radii.

This is a stronger and more nuanced conclusion than the original simple amplitude hypothesis.

8.7 What Was Found About Shell Radius, Group, and Block

The paper found three levels of structure.

First, the major radius definitions occupy reproducible relative positions. Empirical and covalent radii cluster closely, calculated radii occupy a nearby but more variable scale, and van der Waals radii form a distinct outer family.

Second, the radius range ΔR varies systematically by group. Noble gases, halogens, chalcogens, carbon-family elements, and s-block elements tend to show larger full-band widths, while transition-metal groups show narrower widths.

Third, the strongest structural distinction is by block. With the full radius set included, s/p elements have larger full-envelope amplitude, while d/f elements have smaller full-envelope amplitude. When van der Waals radii are removed, the internal radius-family signal reverses direction, with d/f elements showing the larger internal spread.

This reversal is important. It suggests that s/p elements differ most strongly in their outer interaction envelope, whereas d/f elements retain more internal radius-family complexity after R_{vdW} is removed.

8.8 Interpretation Within the Holland Atom Framework

Within the Holland Atom framework, atomic shells are interpreted as resonance-locked standing modes of the Holland Field Φ, triggered and modulated by nuclear mass loading.

A standing mode is not a line. It possesses:

- a mean radial position,
- an amplitude,
- an inner boundary,
- an outer boundary,
- internal structure.

The empirical findings of this paper are consistent with that interpretation. The various accepted radius definitions may be understood as sampling different regions of a finite radial shell envelope:

- covalent and empirical radii sample bonding and mid-shell regions,
- calculated radii sample electron-density-derived structure,
- metallic radii sample packed-state behaviour,
- van der Waals radii sample the outer non-bonded interaction envelope,
- internal structural indicators such as radial nodes may constrain inner shell geometry.

Under this interpretation, the multiplicity of radius values is not a defect in atomic measurement. It is evidence that the shell has depth.

8.9 What the Paper Does Not Yet Prove

The paper does not prove that atomic shells are Holland Field standing modes. It does not prove that A is the exact mechanical amplitude of a physical wave. It does not yet derive A from nuclear mass loading. It also does not prove that each radius type maps uniquely to a fixed position within a standing-wave profile. Those are future requirements.

What this paper does establish is more modest but still significant:

1. Radius variation is structured.
2. Radius variation is statistically organised by periodic-table architecture.
3. Radius variation is not explained by atomic size alone.
4. The full radius envelope contains outer and internal components.
5. The empirical behaviour is compatible with a finite shell-band interpretation.

That is sufficient to justify the next stage of the Holland Atom research programme.

8.10 Nontrivial Findings of the Paper

The most significant findings are:

1. **Atomic radius variation behaves as data, not noise.**
2. **The spread between radius definitions has a characteristic scale.**
3. **The full radius-band half-amplitude is approximately** 38 pm**.**
4. **The amplitude is statistically independent of radius-band midpoint.**
5. **The radius-band structure separates strongly by periodic-table block.**
6. **s/p elements and d/f elements form distinct amplitude populations.**
7. **van der Waals radius contributes strongly to the outer envelope, but block structure survives its removal.**
8. **Removing R_{vdW} reveals a separate internal radius-family signal.**
9. **The results support a two-component shell-envelope interpretation: outer interaction boundary plus internal radius-family spread.**
10. **The findings are consistent with, but do not by themselves prove, the Holland Atom standing-mode shell hypothesis.**

8.11 Overall Interpretation

The paper began with a simple question:

Do multiple atomic radius values merely reflect inconsistent measurement conventions, or do they reveal something physically real about atomic shell structure?

The answer appears to be that they reveal something physically real. The full dataset shows that accepted atomic radius definitions are not randomly dispersed. Their spread is organised, statistically significant, independent of simple atomic-size scaling, and correlated with group and block architecture. The Holland Atom interpretation is therefore plausible: atomic shells may possess finite radial depth, and the various accepted radius definitions may sample different regions within that depth.

- The strongest formulation is not: ***Atomic radius measurements disagree.***
- It is: ***Atomic radius measurements collectively reveal the finite radial envelope of the atomic shell.***

This is the central conceptual result of the paper.

9. Overall Conclusions

The Holland Atom programme began with a fundamental question:

Can the observed architecture of atomic matter be reconstructed from a resonance-lock standing-wave ontology?

Holland Atom Paper 1 demonstrated that a resonance-lock shell framework can reproduce a broad range of observed atomic behaviours, including shell ordering, shell capacities, shell filling sequences, shell contraction, anomalous configurations, and periodic-table structure.

The objective of the present paper was more focused. Before a quantitative shell-geometry scaling law could be developed, it was first necessary to determine which atomic radius definition should be used as the reference geometric radius for the Holland Atom framework. What initially appeared to be a relatively straightforward selection exercise evolved into a much broader investigation of atomic radius variation itself. The results proved unexpectedly informative.

The paper began by establishing that atomic radius is not a uniquely defined quantity. Multiple accepted radius systems coexist throughout physics and chemistry, including empirical atomic radius, calculated atomic radius, covalent radius, metallic radius, ionic radius, and van der Waals radius. Each arises from a different methodology and probes different aspects of atomic structure.

Rather than treating these differences as measurement disagreement, this study examined whether the variation between accepted radius definitions contains physical information. The resulting analyses demonstrated that it does.

The various radius definitions occupy reproducible positions relative to one another, exhibit systematic scaling relationships, and display statistically significant organisation throughout the periodic table. The spread between accepted radius values behaves as a structured observable rather than random variation. The principal empirical findings may be summarised as follows. The radius range,

$$\Delta R = R_{max} - R_{min}$$

was found to possess a characteristic scale throughout the periodic table, with a mean value of approximately:

$$75 \text{ pm}$$

corresponding to a mean half-range amplitude of:

$$A \approx 38 \text{ pm}$$

The radius range exhibits strong periodic dependence and separates into statistically distinct block populations. Using the complete radius dataset, s- and p-block elements exhibit substantially larger radius-band amplitudes than d- and f-block elements. This separation is highly significant statistically and demonstrates that radius variation is strongly linked to atomic architecture. Most importantly, the shell-band amplitude was found to be essentially independent of shell-band midpoint:

$$r(A, R_{mid}) = -0.028 \text{ with } p = 0.785$$

This result eliminates the simplest geometric explanation for radius variation, namely that larger atoms merely possess proportionally larger measurement spreads. The width of the radius band therefore behaves as an independent structural quantity. This is arguably the most significant empirical result obtained in the study.

The robustness analysis presented in Appendix G provided an additional important insight. Removal of van der Waals radius substantially reduces the observed shell-band width but does not eliminate the periodic signal. The amplitude of the full observable shell envelope is effectively independent of atomic size across the complete seven-radius-family dataset, whereas the internal radius family alone shows a significant positive correlation with atomic size (Appendix G). The statistical structure survives, although its character changes. This indicates that atomic radius variation contains at least two components:

1. An outer interaction-envelope component strongly influenced by van der Waals behaviour.
2. An internal radius-family component associated with empirical, calculated, covalent, metallic, and bonding-derived radii.

The **radius variation therefore appears richer and more structured than originally anticipated**. Rather than weakening the interpretation, this result suggests that atomic shell structure may contain multiple observable radial regimes.

The practical objective of the paper was also achieved. Following assessment of the major radius systems, empirical atomic radius was selected as the standard geometric radius for the Holland Atom framework:

$$R_{HA} = R_{empirical}$$

This radius occupies a stable position near the centre of the observed shell band, possesses strong experimental grounding, exhibits robust periodic behaviour, and provides the most appropriate reference geometry for future Holland Atom studies. This selection establishes a consistent geometric foundation for subsequent work. Beyond the practical objective, however, the study has generated several non-trivial observations that appear absent from the existing literature.

To the author's knowledge, no previous study has:

- treated atomic radius variation itself as a measurable observable,

- quantified the periodic-table structure of radius-band width,
- demonstrated statistically distinct block populations in radius variation,
- shown that radius-band amplitude is independent of atomic size,
- or investigated the robustness of radius-band behaviour following systematic removal of van der Waals radii.

These observations emerge directly from accepted experimental and calculated datasets and do not depend upon adoption of the Holland Atom interpretation. They therefore represent independent empirical findings arising from the analysis itself.

Within the Holland Atom framework, these observations acquire a natural interpretation. The Holland Atom proposes that atomic charge shells are resonance-locked standing modes generated through nuclear mass loading of the Holland Field (Φ). Such structures are expected to possess finite radial extent rather than infinitesimally thin boundaries. Under this interpretation, the various accepted radius definitions do not represent conflicting measurements of atomic size. Instead, they represent observations, calculations, and deductions that probe different regions within a finite shell envelope.

The observed radius variation may therefore be interpreted as evidence that atomic shells possess measurable radial depth.

The characteristic shell-band width identified throughout this study is consistent with the existence of such finite shell structure. The standing-mode amplitude hypothesis proposed in Section 5 remains an interpretation rather than a proof. Nevertheless, the empirical evidence presented here demonstrates that atomic radius variation is organised, measurable, statistically significant, and physically meaningful. The possibility that this variation reflects finite shell depth therefore warrants further investigation.

This paper serves as the bridge between Holland Atom Paper 1 and the next phase of the research programme. Paper 1 established the existence of a resonance-lock shell architecture. The present work establishes the geometric reference framework required for quantitative development.

The next paper in the series will use the empirical atomic radius dataset established here as the foundation for deriving the Holland Atom shell-geometry scaling law and associated charge-density scaling relationships.

In doing so, it will test whether the geometric structure identified in this study can be quantitatively predicted from the underlying resonance-lock mechanism.

Regardless of the ultimate interpretation, the results presented here demonstrate that atomic radius variation contains substantially more information than is commonly assumed. What has traditionally been treated as a collection of competing radius definitions may instead represent a measurable window into the radial structure of the atom itself.

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Appendix A. Atomic Radius Definitions and Measurement Methods

A1. Introduction

Unlike macroscopic objects, atoms possess no universally accepted physical boundary. Consequently, multiple radius definitions have been developed for different experimental, theoretical, and practical applications. These radius definitions are not generally equivalent and often produce different numerical values for the same element. The differences arise because each method probes a different aspect of atomic structure, bonding behaviour, electron distribution, or interatomic interaction. This appendix provides a comprehensive baseline catalogue of the principal atomic radius definitions used throughout chemistry and physics [8-16].

Radius Type	Symbol	Standard Definition	Measurement Method	What Is Actually Measured	Typical Use
Empirical Atomic Radius	**Ra**	Radius assigned to a **neutral atom** based on **observed** atomic dimensions and periodic trends	Derived from experimental atomic size observations	Effective size of a neutral atom	General atomic size comparisons
Calculated Atomic Radius	**Rcalc**	Radius obtained from **theoretical electron-density** calculations	Quantum-mechanical electron-density models	Extent of calculated electron distribution	Theoretical chemistry
Covalent Radius	**Rc**	**Half the bond length** between two identical covalently bonded atoms	**Measured** from bond lengths	Stable bonding distance	Molecular structure
Single-Bond Covalent Radius	**Rc,s**	Covalent radius **derived** from single bonds only	Bond-length measurements	Preferred single-bond separation	Organic and inorganic chemistry
Double-Bond Covalent Radius	**Rc,d**	Covalent radius **derived** from double bonds	Bond-length measurements	Reduced bond separation	Bond-order studies
Triple-Bond Covalent Radius	**Rc,t**	Covalent radius **derived** from triple bonds	Bond-length measurements	Maximum bond compression	Bond-order studies
Empirical Covalent Radius	**Rc,e**	Historical covalent radius estimate **derived** from experimental observations	Bond-length surveys	Effective bonding size	Structural chemistry
Metallic Radius	**Rm**	Half the distance between nearest-neighbour atoms in a metallic crystal	Crystal lattice **measurements**	Atomic packing distance in metals	Solid-state physics
Ionic Radius	**Ri**	Effective radius assigned to an ion within a crystal according to a specified radius convention	Crystal structure **analysis**	Apparent size of charged atom	Ionic compounds
van der Waals Radius	**RvdW**	Effective radius inferred from observed non-bonded atomic separations	Molecular crystal **measurements**	Non-bonded interaction limit	Intermolecular interactions
Bohr Radius	**a_0**	Characteristic length scale of hydrogen ground state	**Derived** from atomic theory	Hydrogen orbital scale parameter	Quantum mechanics
Expectation Radius	**$\langle r \rangle$**	Mean electron distance from nucleus	Wavefunction integration	Average electron location	Quantum mechanics
RMS Radius	**$\sqrt{\langle r^2 \rangle}$**	Root-mean-square electron distance from nucleus	Wavefunction integration	Statistical spread of electron distribution	Quantum mechanics
Electron Density Contour Radius	**Rρ**	Radius enclosing a specified electron-density threshold	Density integration	Selected probability-density boundary	Computational chemistry

Table A1. Atomic Radius Definitions

Table A1 contains a summary of the principal atomic radius systems used in this study. Although each radius type is widely accepted within its respective domain, they frequently assign different values to the same element because they probe different physical phenomena, bonding environments, or interaction boundaries. This variation forms the foundation of the shell-band analysis presented in the Holland Atom framework.

Radius Family	Symbol	Primary Source	Basis	Classification
Atomic Radius (Empirical)	Ra	WebElements (derived from experimental atomic dimensions)	Experimental / observational	Neutral atom radius
Atomic Radius (Calculated)	Ra-c	WebElements	Quantum-mechanical electron-density calculations	Theoretical radius
Covalent Radius (2008)	Rc2008	Cordero et al. (2008), *Covalent Radii Revisited*	Crystallographic bond-length analysis	Modern covalent radius
Covalent Radius (Single Bond)	Rc,s	WebElements	Single-bond internuclear distance / 2	Bonding radius
Covalent Radius (Triple Bond)	Rc,t	WebElements	Triple-bond internuclear distance / 2	Bonding radius
Metallic Radius	Rm	WebElements	Metallic crystal nearest-neighbour spacing / 2	Crystal-packing radius
van der Waals Radius	RvdW	WebElements (Bondi / Alvarez compilations)	Non-bonded atomic contact distance / 2	Interaction-boundary radius

Table A2. Radius Families Used in the Holland Atom Radius Audit

Table A2 shows that together the seven radius families span experimental, theoretical, bonding, crystalline, and interaction-derived measures of atomic size, providing a comprehensive basis for investigating atomic radius variation across the periodic table [19,20].

Atomic Radius (Empirical)

Represents experimentally inferred atomic dimensions and forms the Holland Atom reference radius selected in Section 6.

Atomic Radius (Calculated)

Derived from theoretical electron-density models and included to provide a complete calculated radius reference across the periodic table.

Covalent Radius (2008)

The modern covalent radius dataset of Cordero et al. (2008), derived from crystallographic bond lengths and widely regarded as the current standard covalent-radius compilation.

Covalent Single- and Triple-Bond Radii

Included to investigate whether bond-order influences the observed shell-band structure and to provide additional constraints on internal radius-family variation.

Metallic Radius

Represents the effective radius of atoms within metallic crystal structures and provides an independent crystal-packing measure of atomic size.

van der Waals Radius

Represents the outer non-bonded interaction limit of the atom and forms the largest radius family in most elements. Appendix G demonstrates that this radius contributes substantially to the full observable radius envelope.

Below are definitions of several commonly used terms within atomic radius science.

Neutral Atom

A **neutral atom** is an atom containing equal numbers of positive and negative charges.

$$N_p = N_e$$

where:

- N_p = number of protons
- N_e = number of electrons

A neutral atom therefore possesses zero net electric charge. Examples:

- Hydrogen: 1 proton, 1 electron
- Carbon: 6 protons, 6 electrons
- Neon: 10 protons, 10 electrons

Neutral atoms are distinct from ions, which possess unequal numbers of protons and electrons.

Electron Density

Electron density is a mathematical quantity representing the predicted spatial distribution of electronic charge around an atomic nucleus.

It is usually written as:

$$\rho(\mathbf{r})$$

For a general many-electron atom:

$$\rho(\mathbf{r}) = N\int \mid \Psi(\mathbf{r}, \mathbf{r}_2, \mathbf{r}_3, \ldots, \mathbf{r}_N) \mid^2 \; d\mathbf{r}_2 \, d\mathbf{r}_3 \cdots d\mathbf{r}_N$$

where Ψ is the many-electron wavefunction. For a single-electron system, such as hydrogen, this reduces to:

$$\rho(\mathbf{r}) = \mid \psi(\mathbf{r}) \mid^2$$

Electron density is therefore not measured directly as a physical surface or boundary. Instead, it is a theoretical construct derived from a mathematical model of the electronic state. Different quantum-mechanical approximations may produce different electron-density distributions, including:

- Hartree–Fock
- Density Functional Theory (DFT)
- Configuration Interaction
- Coupled Cluster methods

Consequently, radii derived from electron-density calculations are model-dependent.

Bond Length

Bond length is the distance between the nuclei of two bonded atoms. It is typically measured experimentally using:

X-ray Crystallography

Determines atomic positions within crystals from diffraction patterns.

Neutron Diffraction

Uses neutron scattering to determine nuclear positions.

Electron Diffraction

Uses electron scattering from ordered structures.

Microwave and Rotational Spectroscopy

Determines internuclear separations from molecular rotational spectra. Bond lengths are therefore experimentally measured quantities. Atomic radii derived from bond lengths are subsequently obtained by calculation, usually by assigning a fraction of the measured internuclear distance to each atom. For example:

$$R_c = \frac{d_{A-A}}{2}$$

where:

- d_{A-A} is experimentally measured
- R_c is subsequently derived

Thus, covalent radii are not directly measured, but are derived from directly measured bond distances.

A2. Evidence Classification Guide

To distinguish between different forms of scientific evidence, the following terminology is used throughout this paper.

Evidence Class	Definition
Directly Measured	Obtained directly from experimental observation with minimal interpretive processing.
Observationally Inferred	Inferred from experimental observations using accepted empirical conventions or relationships.
Model-Derived	Calculated from experimentally measured quantities through a defined model, approximation, or allocation method.
Theory-Derived	Obtained primarily from theoretical calculations whose values depend upon an underlying physical model.

Evidence Class	Definition
Pure Theoretical Parameter	A mathematical quantity defined by theory and not directly measured experimentally.

Table A3. Evidence Classification key

Table A3 shows that not all atomic radius definitions are established in the same manner. Some are based directly upon experimental measurements, whereas others are inferred from measured data, derived through accepted conventions, or calculated from theoretical models. Table A2 classifies each radius type according to its evidential basis and degree of separation from direct experimental observation. The purpose of this classification is not to rank the scientific validity of a radius definition, but rather to distinguish how each quantity is obtained and what assumptions are required in its determination.

The distinction between *Empirical Covalent Radius (Rc,e)* and *Covalent Radius (Rc)* is one of methodology rather than underlying data source. Both originate from experimentally observed bond lengths. However, empirical covalent radii are historical estimates derived directly from observed chemical and structural behaviour, whereas modern covalent radii are typically obtained using formalised radius models, statistical analyses, and standardised bond-length conventions. Consequently, both Rc,e and Rc are classified here as Model-Derived quantities. The distinction lies in methodology rather than evidential category, with Rc,e representing historical bond-partitioning approaches and Rc representing modern standardised radius models.

Therefore, throughout this paper, the terms: **measured**; **observed**; **derived**; **calculated**; and **theoretical**, are not used interchangeably.

Radius Type	Evidence Class
Empirical Atomic Radius (Ra)	*Observationally Inferred*
Calculated Atomic Radius (Rcalc)	*Theory-Derived*
Covalent Radius (Rc)	*Model-Derived*
Single-Bond Covalent Radius (Rc,s)	*Model-Derived*
Double-Bond Covalent Radius (Rc,d)	*Model-Derived*
Triple-Bond Covalent Radius (Rc,t)	*Model-Derived*
Empirical Covalent Radius (Rc,e)	*Model-Derived*
Metallic Radius (Rm)	*Model-Derived*
Ionic Radius (Ri)	*Model-Derived*
van der Waals Radius (RvdW)	*Observationally Inferred*
Bohr Radius (a_0)	*Pure Theoretical Parameter*
Expectation Radius ($\langle r \rangle$)	*Pure Theoretical Parameter*
RMS Radius ($\sqrt{\langle r^2 \rangle}$)	*Pure Theoretical Parameter*
Electron Density Contour Radius ($R\rho$)	*Theory-Derived*

Table A4. Radius Type vs. Evidence Class

Table A4 defines each radius type as one of the defined categories. Note that none of the radius definitions in Table A1 are actually "Directly Measured". Even covalent radius, ionic radius, metallic radius, and van der Waals radius are derived from measured distances rather than measured directly as radii.

A3. Radius Definitions Based on Chemical Bonding

Covalent Radius

Definition: Half the distance between two identical atoms joined by a covalent bond.

Mathematical Form

$$R_c = \frac{d_{A-A}}{2}$$

where:

- d_{A-A} = measured bond length

Interpretation: Represents the size of an atom during stable covalent bonding.

Single-Bond Covalent Radius

Uses only single-bond measurements. Examples:

- C–C in diamond
- H–H in H_2

Provides the largest covalent radius for a given element.

Double-Bond Covalent Radius

Derived from double bonds. Examples:

- C=C
- O=O

Typically, smaller than single-bond radius due to reduced bond distance.

Triple-Bond Covalent Radius

Derived from triple bonds. Examples:

- N≡N
- C≡C

Usually produces the smallest covalent radius.

A4. Radius Definitions Based on Crystal Structure

Metallic Radius

Definition: Half the nearest-neighbour separation in a metallic lattice.

Formula

$$R_m = \frac{d_{NN}}{2}$$

Interpretation: Represents effective atomic size within metallic packing.

Ionic Radius

Definition: Effective size assigned to an ion within an ionic crystal.

Determination: Ionic radii are obtained indirectly from crystal structure measurements.

Experimental techniques such as X-ray diffraction and neutron diffraction determine the distances between neighbouring ions within a crystal lattice. Individual ionic radii are then assigned by partitioning the measured ion–ion separation according to an accepted radius model.

Important Characteristics: Ionic radius depends upon:

- charge state
- coordination number
- crystal environment
- radius partitioning convention

For example:

- Fe^{2+} and Fe^{3+} possess different ionic radii.
- Six-coordinate and eight-coordinate ions typically possess different assigned radii.
- Different radius systems may assign slightly different values to the same ion.

Radius Conventions

Several ionic-radius systems have been proposed, including those of Goldschmidt, Pauling, and Shannon [7]. The most widely used modern compilation is the Shannon ionic radius system, which incorporates coordination number and oxidation state effects. Because ionic radii are assigned rather than directly measured, absolute values may differ between radius conventions [11]. Differences of approximately 10–20 pm can occur for some ions when comparing historical datasets.

Interpretation: Ionic radius represents an effective size parameter that reproduces observed crystal geometries and interionic separations. It should therefore be regarded as a model-derived quantity rather than a directly measured physical boundary.

A5. Radius Definitions Based on Non-Bonded Interactions

van der Waals Radius

Definition: The effective radius of an atom derived from the closest approach of non-bonded atoms.

It is commonly expressed as one-half of the observed separation between two non-bonded atoms:

$$R_{vdW} = \frac{d_{non-bonded}}{2}$$

where:

- $d_{non-bonded}$ is the experimentally observed separation between neighbouring non-bonded atoms.

Determination: van der Waals radii are obtained indirectly from observations of non-bonded atomic interactions. Sources used in radius compilations include:

- molecular crystal packing measurements
- X-ray crystallography
- neutron diffraction
- gas-phase scattering experiments
- spectroscopic measurements
- computational modelling

Most modern datasets combine information from multiple sources rather than relying upon a single experimental method.

Radius Compilations: Several widely used van der Waals radius compilations exist, including those of:

- Bondi (1964)
- Rowland and Taylor (1996)
- Mantina et al. (2009)
- Alvarez (2013)

Because different datasets utilise different source information and fitting procedures, published van der Waals radii may vary slightly between compilations [8,16].

Interpretation: van der Waals radius represents an effective interaction boundary for atoms that are not chemically bonded. It is often interpreted as the approximate outer limit of an atom's influence during weak intermolecular interactions and molecular packing.

Important Characteristics: van der Waals radius depends upon:

- the source dataset used
- experimental methodology
- interpretation of non-bonded contact distances
- compilation convention

Consequently, van der Waals radii should be regarded as observationally inferred quantities rather than directly measured physical boundaries.

A6. Radius Definitions from Quantum Theory

Calculated Atomic Radius

Definition: Radius obtained from theoretical electron-density calculations.

Determination: Computed using:

- Hartree–Fock methods
- Density Functional Theory (DFT)
- Post-Hartree–Fock methods

Interpretation: Represents the extent of the calculated electron distribution.

Expectation Radius

Definition: The expectation radius is the mean radial distance of the electron probability distribution from the nucleus:

$$\langle r \rangle = \int r \ | \psi |^2 \ dV$$

or equivalently, for a spherically symmetric radial distribution:

$$\langle r \rangle = \int_0^\infty r \, P(r) \, dr$$

where $P(r)$ is the radial probability density.

Interpretation: Represents the average distance of the electron distribution from the nucleus.

Root-Mean-Square (RMS) Radius

Definition: The RMS radius is the square root of the second moment of the electron distribution:

$$r_{\text{RMS}} = \sqrt{\langle r^2 \rangle} = \sqrt{\int r^2 \ | \psi |^2 \ dV}$$

or equivalently:

$$r_{\text{RMS}} = \sqrt{\int_0^\infty r^2 \, P(r) \, dr}$$

Interpretation: Represents the characteristic radial extent of the electron distribution and is sensitive to the outer tails of the distribution by giving greater weight to probability density located farther from the nucleus. Because larger radii are weighted more heavily through the r^2 term, $r_{\text{RMS}} \geq \langle r \rangle$, for any non-trivial distribution.

Electron-Density Contour Radius

Definition: Radius enclosing a chosen electron-density threshold.

Common thresholds include:

- 95% electron density
- 98% electron density
- 99% electron density

Interpretation: Defines a practical boundary for electron distribution.

A7. Radius Definitions from Atomic Theory

Bohr Radius

Definition: Characteristic radius of the hydrogen ground state.

$$a_0 = 5.29177210903 \times 10^{-11}\ \text{m}$$

Historical Importance: Serves as the fundamental length scale of the Bohr model and remains widely used in atomic physics.

Interpretation: Theoretical scale length associated with hydrogen.

A8. Important Observations

Observation A1

There is no universally accepted atomic boundary.

Observation A2

Different radius definitions routinely produce different numerical values for the same element.

Observation A3

Most radius measurements are indirect and are inferred from bonding, crystal structure, interaction distances, or theoretical calculations.

Observation A4

The radius assigned to an atom depends on the physical phenomenon being investigated.

Observation A5

Atomic radius should therefore be regarded as a family of related measurements rather than a single uniquely defined quantity.

Observation A6

No commonly used atomic radius is directly measured as a physical radius. All accepted radius definitions are inferred, derived, assigned, or calculated from other experimentally observed quantities.

Appendix B part 1. Radius Variation in Carbon

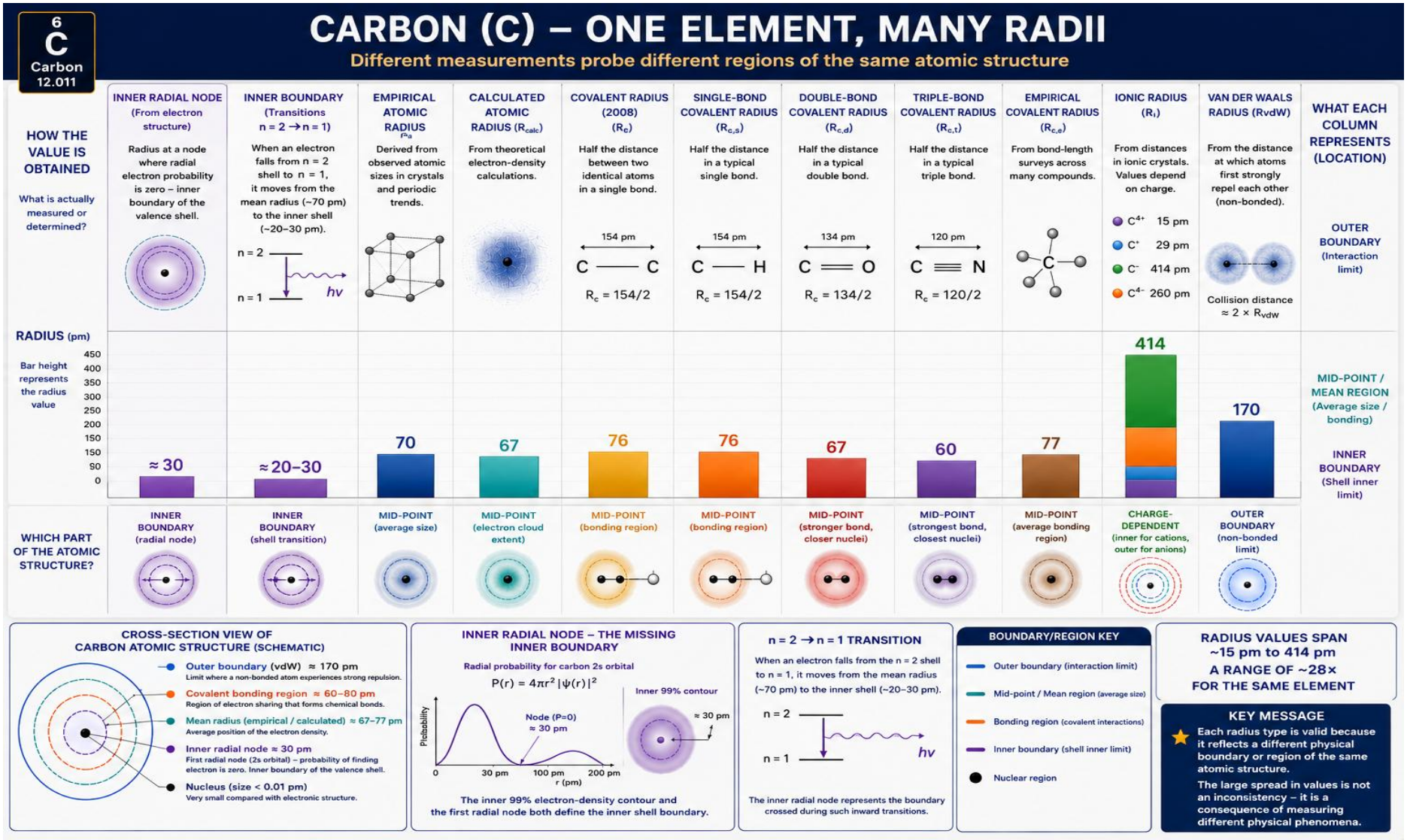


Figure B1. Carbon (C): One Element, Many Radii.

Appendix B Part 2. Carbon Radius Variation: Analysis, Interpretation and Preliminary Conclusions

B2.1 Purpose

Appendix B Part 1 established a detailed radius inventory for Carbon using the radius definitions introduced in Appendix A. Carbon was selected because it possesses one of the most complete sets of radius measurements available for any element, including empirical, calculated, covalent, ionic and non-bonded radius definitions. The objective of this appendix is to examine the relationships between these radius values, identify any systematic ordering that emerges, and assess whether the various radius definitions may be probing different regions of a common underlying atomic structure. No Holland Atom interpretation is introduced at this stage. The analysis remains strictly observational.

B2.2 Summary of Carbon Radius Dataset

Radius Type	Physical State
Empirical Atomic Radius	Neutral Carbon atom (derived/inferred atomic dimension)
Calculated Atomic Radius	Neutral isolated Carbon atom (theoretical electron density)
Covalent Radius (2008)	Carbon in bonded compounds (average crystallographic environments)
Covalent Single Bond	Carbon participating in a C–C single bond
Covalent Double Bond	Carbon participating in a C=C double bond
Covalent Triple Bond	Carbon participating in a C≡C triple bond
Metallic Radius	Not really applicable to ordinary Carbon; often omitted or estimated
Ionic Radius	Specific ionic state (e.g. C^{4-} where available)
van der Waals Radius	Non-bonded Carbon interaction distance

Table B-P2.1. Radius Systems Relating to Carbon State and Bonding

The Carbon case study assembled the following accepted radius values from the literature.

Radius Type	Symbol	Radius (pm)
Inner Radial Node (structural indicator)	Rnode	~30
Empirical Atomic Radius	Ra	70
Calculated Atomic Radius	Rcalc	67
Covalent Radius (2008)	Rc	76
Single-Bond Covalent Radius	Rc,s	76
Double-Bond Covalent Radius	Rc,d	67
Triple-Bond Covalent Radius	Rc,t	60
Empirical Covalent Radius	Rc,e	77
van der Waals Radius	RvdW	170

Table B-P2.2. Carbon Radius Values Across Different Radius Systems

The inner radial node is not treated as an atomic radius measurement. It is included as an internal structural indicator that may constrain the inner extent of the shell band. Additional ionic radius values were identified but were excluded from the primary hierarchy analysis because ionic radii are strongly dependent upon charge state and crystal environment.

B2.3 Radius Variation Exists Within a Single Element

Carbon provides one of the most complete collections of published atomic radius measurements available for any element. The accepted radius systems examined in this study span values from approximately 60 pm (triple-bond covalent radius) to 170 pm (van der Waals radius), representing a range of almost a factor of three within a single atomic species. This result is significant because all of these values are accepted within their respective experimental or theoretical contexts. The variation therefore does not arise from disagreement regarding the identity of the element, but from different methodologies probing different aspects of atomic structure and atomic interaction.

The lower end of the accepted radius family is represented by bonding-derived radii, particularly triple-bond and double-bond covalent radii. Intermediate values are occupied by empirical atomic radius, calculated atomic radius, and covalent radius measurements. The largest accepted radius is the van der Waals radius, which defines the outer non-bonded interaction boundary at which neighbouring atoms first experience significant repulsive interaction.

In addition to these accepted radius measurements, the radial probability distribution of Carbon contains an inner radial node located at approximately 30 pm. This quantity is not treated as an atomic radius measurement and is therefore excluded from the accepted radius family used throughout the statistical analyses presented in this paper. Instead, it is included as an internal structural indicator because it identifies a physically meaningful boundary within the shell architecture itself.

Consequently, two distinct radial spans may be identified:

Accepted Radius Envelope

- Triple-bond covalent radius: ~60 pm
- van der Waals radius: ~170 pm
- Range ratio: ~2.8×

This envelope represents the range spanned by accepted radius definitions currently used throughout chemistry and atomic physics.

Extended Structural Envelope

- Inner radial node: ~30 pm
- van der Waals radius: ~170 pm
- Range ratio: ~5.7×

This larger span includes both the innermost structural indicator and the outermost interaction boundary, providing an estimate of the full radial domain associated with the Carbon shell structure.

The distinction between these two envelopes is important. The accepted radius envelope is entirely observational and forms the basis of the statistical analyses conducted throughout this paper. The extended structural envelope incorporates additional information regarding potential internal and external shell boundaries. Within the Holland Atom interpretation, this broader radial span is consistent with the possibility that atomic shells possess finite radial extent rather than existing as infinitesimally thin boundaries.

The Carbon dataset therefore demonstrates two important observations. First, substantial radius variation exists even within a single element. Second, the radius values exhibit a clear and reproducible hierarchy extending from internal structural indicators, through bonding and empirical

radii, to outer interaction boundaries. This ordered behaviour motivates the subsequent investigation of whether similar structure exists throughout the periodic table.

B2.4 Internal Consistency of the Covalent Radius Family

One of the most striking observations emerges from the covalent radius measurements.

Bond Type	Radius (pm)
Triple Bond	60
Double Bond	67
Single Bond	76

Table B-P2.3. Carbon Radius Values by Bond Type

A clear monotonic progression is observed.

As bond order increases: Single → Double → Triple
the assigned radius decreases: 76 → 67 → 60 pm

Conversely, increasing bond strength produces systematically smaller assigned radii. This behaviour is not arbitrary. Each radius is obtained from: R = Bond Length / 2. Consequently, stronger bonding interactions correspond to shorter internuclear separations and therefore smaller assigned radii. The covalent radius family therefore exhibits an internally coherent structure that reflects measurable physical changes in bonding behaviour.

B2.5 Clustering of Empirical and Calculated Radii

A second notable observation is the close agreement between the empirical and calculated radius definitions.

Radius Type	Radius (pm)
Calculated Atomic Radius	67
Empirical Atomic Radius	70
Covalent Radius (2008)	76
Empirical Covalent Radius	77

Table B-P2.4. Carbon Radius Variation by Radius Type

These values occupy a relatively narrow interval: 67–77 pm representing a spread of only approximately ±7% around the central value. Despite being derived from different methodologies:

- experimental observation
- electron-density calculations
- bond-length measurements
- historical compilations

all four radius definitions converge upon a common radial region. This convergence suggests that these methods are probing closely related aspects of atomic structure.

B2.6 Separation of van der Waals Radius

The van der Waals radius differs markedly from the remaining radius definitions, with Carbon displaying a van der Waals Radius of 170 pm. This value is approximately: 170 / 70 ≈ 2.43 times larger than the empirical atomic radius. Unlike the covalent and empirical radius families, the van der Waals radius represents the distance at which non-bonded atoms first experience strong repulsive interaction. It therefore appears to probe a substantially more external region of atomic structure.

B2.7 Emergence of a Three-Region Hierarchy

When the Carbon radius values are arranged in ascending order, a distinct hierarchy emerges.

Region	Representative Values
Inner Region	~30 pm
Intermediate Region	60–77 pm
Outer Region	170 pm

Table B-P2.5. A Carbon Radius Hierarchy Emerges

The intermediate region contains:

- Triple-Bond Radius
- Double-Bond Radius
- Calculated Radius
- Empirical Radius
- Single-Bond Radius
- Empirical Covalent Radius

These values form a tightly clustered family. The van der Waals radius forms a clearly separated outer boundary. The radial node occupies a distinct inner position. This arrangement suggests the existence of three physically meaningful radial domains:

1. An inner boundary region.
2. An active interaction region.
3. An outer interaction boundary.

The appearance of these domains arises directly from the measured radius values and does not depend upon any specific theoretical model.

B2.8 Significance of the Inner Radial Node

The addition of the inner radial node introduces a quantity fundamentally different from conventional radius definitions. Carbon exhibits a factor-of-2.3 variation across accepted radius definitions. Inclusion of the first internal radial node extends the observable shell-structure span to approximately a factor of 5.7.

Unlike empirical, covalent or van der Waals radii, the radial node corresponds to a location at which radial electron probability falls to zero. Consequently, the node represents a natural internal boundary rather than an external extent.

This distinction is important because the overwhelming majority of atomic radius definitions catalogue external dimensions, interaction distances or mean radial positions. Very few quantities

attempt to characterise internal shell boundaries. The radial node therefore provides a potentially useful reference point for understanding how different radius definitions relate to distinct regions within atomic structure.

B2.9 Preliminary Conclusions

The Carbon case study establishes several important observations.

1. Multiple accepted radius values coexist for the same element.
2. Radius variation is systematic rather than random.
3. The covalent radius family exhibits a consistent ordering with bond order.
4. Empirical, calculated and covalent radius definitions cluster within a narrow radial interval.
5. The van der Waals radius occupies a distinct outer region.
6. The radial node occupies a distinct inner region.
7. The complete dataset suggests a hierarchical organisation of radius definitions extending from an inner boundary, through an intermediate interaction region, to an outer interaction boundary.

These findings provide the first indication that different radius definitions may be probing different radial zones of the same atomic structure rather than describing entirely independent physical entities.

The next appendix extends this analysis from a single-element case study to a broader statistical comparison across multiple elements and radius systems.

Appendix C. Comparative Analysis of Atomic Radius Definitions

C1. Objective

Appendix A established that multiple atomic radius definitions are used throughout chemistry and physics, and that these definitions frequently assign different numerical values to the same element. The purpose of Appendix C is to investigate whether these differing radius definitions exhibit systematic relationships when compared against a common reference radius.

A representative sample of ten elements was selected spanning:

- hydrogen
- noble gases
- non-metals
- semiconductors
- alkali metals
- transition metals
- heavy elements

The selected operational reference radius is the **Empirical Atomic Radius (Ra)**, which forms the basis of the Paper 2 scaling-law investigation. All alternative radius definitions are therefore expressed as a ratio relative to:

$$R_a = 1.00$$

This appendix is purely observational and introduces no Holland Atom interpretation.

C2. Representative Sample Dataset

The sample elements are H, He,C, O, Na, Si, Fe, Cu, Xe, and U. The selected radius definitions are:

- Empirical Atomic Radius
- Calculated Atomic Radius
- Covalent Radius (2008)
- Empirical Covalent Radius
- van der Waals Radius

Ionic radii, metallic radii and orbital radii are excluded from the primary comparison because they are highly dependent upon charge state, coordination environment, crystal structure, or orbital assignment and therefore do not represent a single radius value for a neutral atom.

C3. Normalisation Method

For each available radius value:

$$R^* = \frac{R}{R_a}$$

where:

- R = radius under investigation
- R_a = Empirical Atomic Radius

Thus, $R_a = 1.00$ for every element.

Radius Type	H	C	O	Na	Si	Fe	Cu	U
Atomic Radius (empirical)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Atomic Radius (calculated)	2.12	—	—	—	—	—	—	0.62
Covalent Radius (2008)	1.24	1.09	1.10	0.92	1.01	0.94	0.98	1.12
Covalent Radius (empirical)	1.28	1.10	1.22	0.86	1.01	0.89	1.02	0.74
van der Waals Radius	4.80	2.43	2.55	1.39	1.99	1.74	1.76	1.74

Table C1. Normalised Radius Ratios Relative to Empirical Atomic Radius

C4. Mean Radius Relationships

Using all available values:

Radius Type	Mean Ratio
Atomic Radius (empirical)	1.00
Covalent Radius (2008)	1.05
Covalent Radius (empirical)	1.01
van der Waals Radius	2.30

Table C2. Mean Ratios Relative to Empirical Atomic Radius

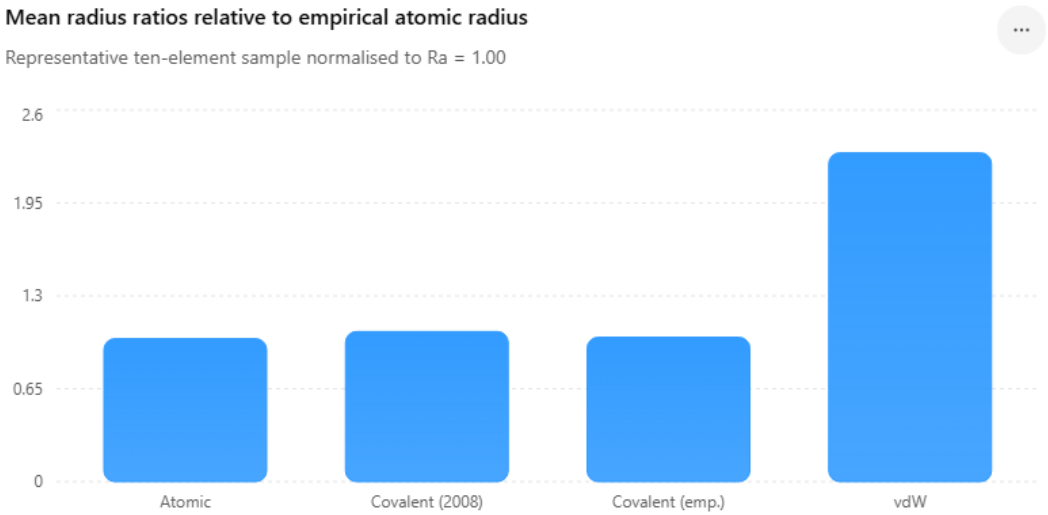


Figure C1. Mean Normalised Radius Ratios

Mean radius ratios relative to empirical atomic radius. Representative ten-element sample normalised to Ra = 1.00

C5. Observations

Observation C1: Not all radius definitions are available for all elements.

Examples include:

- Helium lacks an empirical atomic radius assignment.
- Xenon lacks an empirical atomic radius assignment.
- Hydrogen lacks a metallic radius assignment.
- Many elements lack unique ionic radius assignments.

The existence of a radius value therefore depends partly upon the measurement methodology employed.

Observation C2: The two covalent-radius systems closely track the empirical atomic radius.

Mean values are:

$$R_c(2008) = 1.05R_a$$

and

$$R_c(empirical) = 1.01R_a$$

respectively. This suggests that both covalent-radius systems describe dimensions very similar to the empirical atomic radius.

Observation C3: The van der Waals radius is consistently larger than all other radius definitions examined. Across the representative sample:

$$R_{vdW} \approx 2.3R_a$$

on average. For every element where data are available:

$$R_{vdW} > R_a$$

Observation C4

Hydrogen behaves as a clear outlier. For hydrogen:

$$\frac{R_{vdW}}{R_a} = 4.8$$

which is substantially larger than the corresponding ratios observed for heavier elements. Hydrogen may therefore require separate treatment when comparing radius systems.

Observation C5: The ratio between van der Waals radius and empirical atomic radius appears more consistent among heavier elements than the raw radius values themselves.

For the heavier elements examined:

$$1.7 \lesssim \frac{R_{vdW}}{R_a} \lesssim 2.0$$

despite substantial variation in absolute atomic size.

C6. Preliminary Conclusions

The representative sample demonstrates that different radius definitions do not vary randomly.

Instead, systematic relationships appear to exist between the principal radius systems used throughout chemistry.

In particular:

1. Covalent radii closely track empirical atomic radii.
2. van der Waals radii are consistently larger than empirical atomic radii.
3. Certain radius definitions are unavailable for specific classes of elements.
4. The relative ordering of radius types appears broadly reproducible across chemically diverse elements.

These observations suggest that the various radius definitions may be related through a common underlying structural hierarchy rather than representing entirely independent measures of atomic size.

The next appendix therefore examines how current atomic theory interprets the existence of multiple radius definitions and the differing values they produce.

Appendix D. Full Periodic-Table Radius Scaling Audit

D1. Objective

Appendix D extends the single-element Carbon analysis and representative sample audit to the full available periodic-table dataset. The objective is to determine whether the principal radius systems exhibit systematic scaling relationships when normalised against the selected reference radius:

$$R_a = \text{Empirical Atomic Radius}$$

The three principal comparison ratios were:

$$R^*_{calc} = \frac{R_{calc}}{R_a}$$

$$R^*_{c2008} = \frac{R_{c2008}}{R_a}$$

$$R^*_{vdW} = \frac{R_{vdW}}{R_a}$$

where:

- R_{calc} = calculated atomic radius
- R_{c2008} = Cordero 2008 covalent radius
- R_{vdW} = van der Waals radius
- R_a = empirical atomic radius

Only valid paired values were used. Missing data were not interpolated or estimated.

D2. Dataset Completeness

Radius Dataset	Available Values	Coverage vs 118 Elements	Missing Values
Empirical Atomic Radius R_a	86	72.9%	32
Calculated Atomic Radius	83	70.3%	35
Cordero Covalent Radius 2008	96	81.4%	22
van der Waals Radius	54	45.8%	64

Table D1. Dataset Completeness

The audit confirms that no single radius system provides complete coverage of the periodic table. This supports the earlier conclusion that atomic radius is not a universally available directly measured property, but a method-dependent assigned, inferred, derived, or calculated quantity.

D3. Global Radius Scaling Results

Ratio	N	Mean	Std. Dev.	Min	Max
R_{calc}/R_a	75	1.122	0.180	0.711	1.430
R_{c2008}/R_a	86	1.029	0.090	0.737	1.240
R_{vdW}/R_a	45	1.549	0.663	1.019	4.800

Table D2. Global mean radius ratios

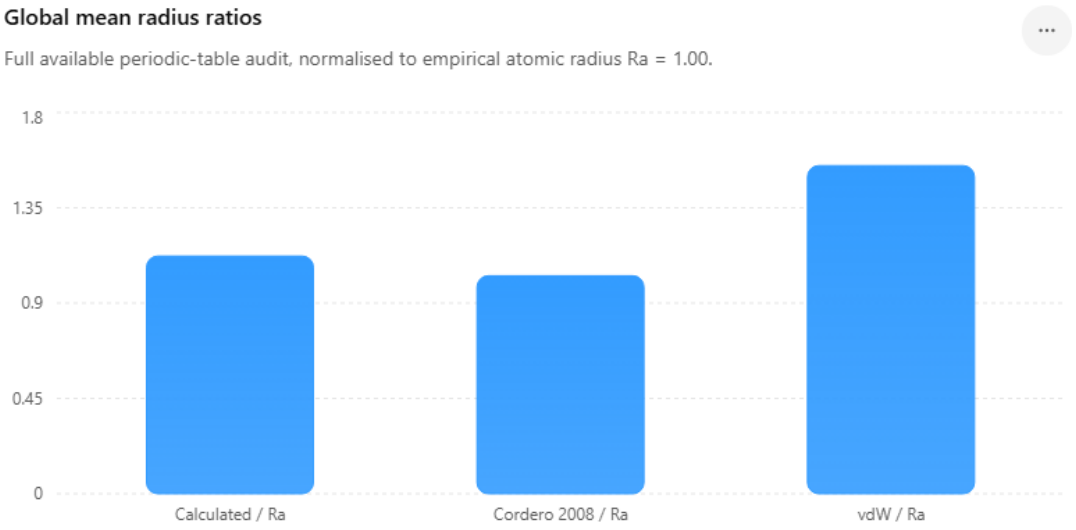


Figure D1. Full periodic-table audit normalised to empirical atomic radius Ra = 1.00.

Interpretation

The Cordero 2008 covalent radius is the closest global match to the empirical atomic radius. Its mean ratio is:

$$R_{c2008}/R_a = 1.029$$

with a relatively small standard deviation of 0.090. This indicates that the Cordero covalent radius and empirical atomic radius occupy nearly the same radial scale across the available dataset. The calculated radius also sits close to the empirical scale on average, but shows greater variation:

$$R_{calc}/R_a = 1.122$$

The van der Waals radius forms a clearly distinct larger-radius family:

$$R_{vdW}/R_a = 1.549$$

The high standard deviation is partly driven by hydrogen, where:

$$R_{vdW}/R_a = 4.8$$

Excluding hydrogen, the mean van der Waals ratio becomes:

$$R_{vdW}/R_a = 1.475$$

D4. Correlation with Empirical Atomic Radius

Radius Type	Paired N	Correlation with R_a
Calculated Atomic Radius	75	0.885
Cordero Covalent Radius 2008	86	0.940
van der Waals Radius	45	0.793

Table D3. Correlation with Empirical Atomic Radius

The strongest relationship is between empirical atomic radius and Cordero 2008 covalent radius. This supports the conclusion that empirical atomic radius and covalent radius are not independent radius families. They appear to describe closely related atomic dimensions.

D5. Group Scaling Results

The group analysis reveals that radius ratios are not uniform across the periodic table.

Selected group means:

Group	R_{calc}/R_a	R_{c2008}/R_a	R_{vdW}/R_a
1	1.117	0.974	1.862
2	1.077	0.972	1.284
13	0.968	0.915	1.489
14	0.992	1.003	1.881
15	0.929	1.016	1.702
16	0.833	0.983	1.699
17	0.817	1.049	1.928

Table D4. Group Scaling Results for Normalised Ratios

Interpretation

The Cordero covalent radius remains close to the empirical atomic radius across most groups. However, the van der Waals ratio varies strongly by chemical family.

The highest group-level van der Waals ratios occur in:

- Group 17 Halogens: 1.928
- Group 14 Carbon group: 1.881
- Group 1 Alkali metals: 1.862
- Group 15 Pnictogens: 1.702
- Group 16 Chalcogens: 1.699

This indicates that van der Waals radius is not merely a constant expansion of empirical radius. It is family-dependent.

D6. Period Scaling Results

Period	R_{calc}/R_a	R_{c2008}/R_a	R_{vdW}/R_a
1	—	1.240	4.800
2	0.957	1.029	2.180
3	0.946	0.997	1.592
4	1.083	0.995	1.380
5	1.136	1.011	1.284
6	1.227	1.044	1.186
7	—	1.086	1.190

Table D5. Period Scaling Results for Normalised Ratios

Interpretation

A strong period trend is visible in the van der Waals ratio.

The mean van der Waals ratio decreases from:

$$2.180$$

in Period 2 to approximately:

$$1.186$$

in Period 6.

This suggests that the relative separation between empirical radius and van der Waals radius is largest for lighter elements and progressively narrows in heavier elements. The Cordero covalent radius remains far more stable across periods, remaining close to:

$$R_{c2008}/R_a \approx 1.0$$

throughout the dataset.

D7. Radius Hierarchy Test

For all elements where empirical, Cordero, and van der Waals radii were all available:

- $N = 45$
- $R_{vdW} > R_a$ in 45/45 cases
- $R_{vdW} > R_{c2008}$ in 44/45 cases
- $R_{vdW} > R_{c2008} > R_a$ in 15/45 cases

The universal result is:

$$R_{vdW} > R_a$$

for every available paired case.

However, the ordering between R_a and R_{c2008} varies by element. This is an important result. It shows that van der Waals radius forms a distinct outer radius family, while empirical and covalent radii occupy a shared intermediate region whose ordering may vary locally.

D8. Outliers

Ratio	Minimum	Maximum
R_{calc}/R_a	Polonium, 0.711	Tungsten, 1.430
R_{c2008}/R_a	Polonium, 0.737	Hydrogen, 1.240
R_{vdW}/R_a	Cadmium, 1.019	Hydrogen, 4.800

Table D6. Population Outliers

Hydrogen is the strongest van der Waals outlier. This is consistent with earlier observations from the Carbon and representative-sample analyses: light elements, especially hydrogen, may require separate treatment when comparing radius definitions.

Polonium represents the most extreme negative outlier in both the $R_{calc}/R_a$and R_{c2008}/R_a normalised datasets. This behaviour is consistent with the well-established influence of relativistic effects in heavy p-block elements, where increasing nuclear charge leads to relativistic contraction and stabilisation of inner electron states, producing measurable deviations from simple periodic radius trends [17]. The observation therefore appears to reflect known heavy-element atomic physics rather than a deficiency in the underlying radius datasets.

D9. Principal Findings

Finding D1

The Cordero 2008 covalent radius closely tracks empirical atomic radius.

$$R_{c2008}/R_a = 1.029 \pm 0.090$$

This indicates that both radius systems occupy a common intermediate radial scale.

Finding D2

Calculated atomic radius is broadly aligned with empirical radius but shows greater variation.

$$R_{calc}/R_a = 1.122 \pm 0.180$$

This is consistent with its theory-derived status.

Finding D3

van der Waals radius forms a distinct outer-radius family.

$$R_{vdW}/R_a = 1.549 \pm 0.663$$

The relationship is systematic but not constant.

Finding D4

Radius relationships are group-dependent.

The strongest van der Waals expansions occur in Groups 1, 14, 15, 16, and 17.

Finding D5

Radius relationships are period-dependent.

The van der Waals-to-empirical ratio decreases strongly from light periods to heavier periods.

Finding D6

The existence of missing data is itself significant.

The audit confirms that radius definitions are not universally available for all elements, reinforcing the conclusion that atomic radius is a measurement-framework-dependent quantity.

D10. Overall Conclusion

The full periodic-table audit confirms that atomic radius variation is not random.

The principal radius systems form reproducible scaling relationships:

- empirical and covalent radii cluster together,
- calculated radii occupy a nearby but more variable scale,
- van der Waals radii form a consistently larger outer family.

The audit therefore supports the central conclusion of the paper:

atomic radius is not a single physical quantity, but a structured family of radius measurements that appear to probe different radial domains of atomic behaviour.

This provides a quantitative foundation for the next stage of the paper: interpreting whether these observed radius families correspond to distinct structural regions within the atom.

Appendix E. Group and Block Dependence of Atomic Radius Variation

E1. Objective

Appendices B–D established that multiple accepted radius definitions exist for individual elements and that the resulting radius variation is systematic rather than random. The periodic-table audit further demonstrated that the spread between accepted radius values forms a measurable quantity that varies significantly between elements. The objective of Appendix E is to determine whether this radius variation exhibits periodic-table structure.

Unlike the baseline audit presented in Sections 2–4, the analyses in Appendix E utilise the complete seven-radius-family dataset (empirical atomic radius, calculated atomic radius, Cordero covalent radius (2008), single-bond covalent radius, triple-bond covalent radius, metallic radius, and van der Waals radius) to investigate the full observable structure of atomic radius variation across the periodic table.

Specifically, the analysis investigates whether the radius range width,

$$\Delta R = R_{\max} - R_{\min}$$

is dependent upon:

- Periodic-table Group (1–18)
- Periodic-table Block (s, p, d, f)

where $R_{\max}$and $R_{\min}$ are respectively the largest and smallest accepted radius values available for a given element.

If radius variation arises solely from measurement uncertainty or inconsistent methodologies, no systematic group or block dependence would be expected. Conversely, statistically significant dependence would indicate that radius variation contains genuine periodic-table structure.

E2. Methodology

For each element within the master dataset, all available radius values were collected from the selected radius systems:

- Empirical Atomic Radius
- Calculated Atomic Radius
- Covalent Radius (Cordero 2008)
- van der Waals Radius
- Covalent Radius (Single Bond)
- Covalent Radius (Triple Bond)
- Metallic Radius

The radius range width was then calculated as:

$$\Delta R = R_{\max} - R_{\min}$$

Only elements possessing at least two valid radius measurements were included in the analysis.

A total of:

$$N = 96$$

elements satisfied this criterion.

The resulting ΔR values were then grouped according to:

1. Periodic-table Group (1–18)
2. Periodic-table Block (s, p, d, f)

Statistical comparisons were performed using:

- Descriptive statistics
- Boxplot visualisation
- One-way ANOVA
- Kruskal–Wallis non-parametric analysis
- Pairwise Welch unequal-variance t-tests

E3. Group Dependence

Figure E1 presents the distribution of radius range width by periodic-table group.

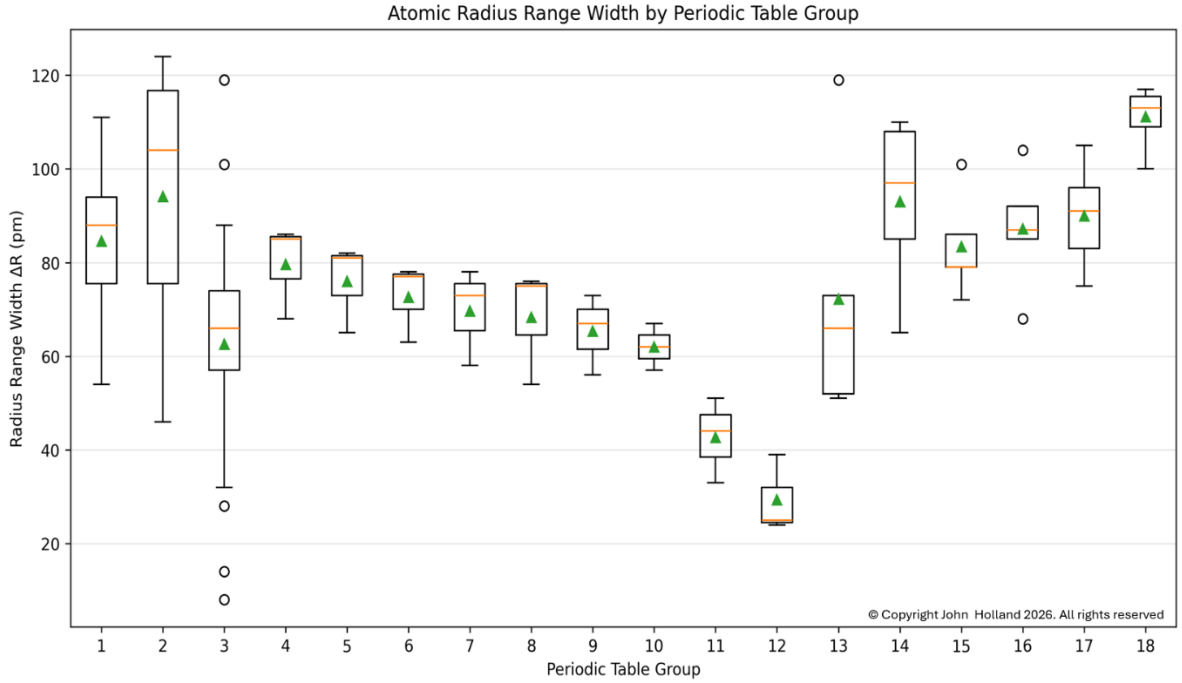


Figure E1. Radius Range Width by Group

Visual inspection immediately reveals substantial variation between groups.

Several groups exhibit consistently large radius ranges, particularly:

- Group 18 (Noble Gases)
- Group 17 (Halogens)
- Group 16 (Chalcogens)
- Group 14 (Carbon Group)
- Groups 1–2 (Alkali and Alkaline-Earth Metals)

Conversely, many transition-metal groups exhibit substantially smaller radius ranges.

The largest group mean occurs within Group 18: $\Delta\bar{R}_{18} = 111.2$ pm

while the smallest occurs within Group 12: $\Delta\bar{R}_{12} = 29.3$ pm

representing a difference of approximately: 82 pm between the two extremes. This variation is considerably larger than would be expected from random measurement scatter alone.

E4. Block Dependence

To investigate whether the observed group behaviour reflects broader periodic-table organisation, the data were grouped according to periodic-table block.

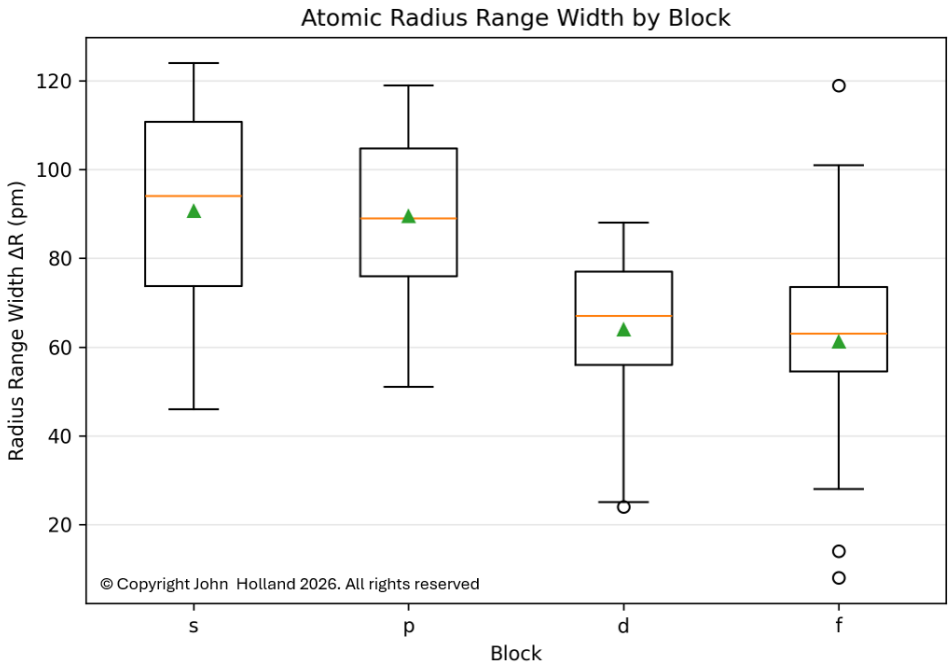


Figure E2. Atomic Radius Range Width by Block.

Figure E2 shows that the block analysis reveals an immediately apparent separation between two populations. This is not random noise or measurement error: this is a signal of deeper underlying mechanism.

Wide-Range Population

Block	Mean ΔR
s	90.64 pm
p	89.47 pm

Narrow-Range Population

Block	Mean ΔR
d	63.97 pm

Block	Mean ΔR
f	61.17 pm

The difference between the two populations is approximately: 27 pm, on average.

The median values show the same behaviour, indicating that the effect is not driven by a small number of outliers.

E5. Statistical Significance

One-Way ANOVA

The null hypothesis was:

$$H_0: \mu_s = \mu_p = \mu_d = \mu_f$$

where μ represents the mean radius range width of each block.

The ANOVA result was:

$$F = 13.10$$
$$p = 3.39 \times 10^{-7}$$

The null hypothesis is therefore rejected. The mean radius range width differs significantly between periodic-table blocks.

Kruskal–Wallis Test

To verify that the result was not dependent upon normality assumptions, a non-parametric Kruskal–Wallis test was also performed.

$$H = 28.98$$
$$p = 2.26 \times 10^{-6}$$

The result remains highly significant.

E6. Pairwise Comparisons

Pairwise Welch unequal-variance t-tests were performed between all block combinations.

The results are summarised below.

Comparison	p-value	Interpretation
s vs p	0.874	Not significant
s vs d	0.0015	**Significant**
s vs f	0.0015	**Significant**
p vs d	1.63×10^{-6}	**Significant**
p vs f	7.35×10^{-5}	**Significant**

Comparison	p-value	Interpretation
d vs f	0.659	Not significant

Two clear observations emerge. First, the s- and p-block populations are statistically indistinguishable from one another. Second, the d- and f-block populations are likewise statistically indistinguishable from one another.

However, both s and p differ strongly from both d and f. This indicates the existence of two statistically distinct populations.

E7. Combined Population Analysis

The block data were therefore combined into two broader populations:

Population A

$$s + p$$

Population B

$$d + f$$

The resulting mean values were:

Population	Mean ΔR
s + p	89.84 pm
d + f	62.73 pm

A Welch unequal-variance comparison yielded: $p = 8.64 \times 10^{-9}$

A Mann–Whitney non-parametric comparison yielded: $p = 7.76 \times 10^{-8}$

Both tests strongly reject the hypothesis that the two populations arise from the same parent distribution.

The effect size was: $d = 1.29$, which constitutes a large effect according to conventional statistical interpretation.

E8. Interpretation

The results demonstrate that radius variation is not randomly distributed throughout the periodic table.

Instead, the radius range width exhibits strong dependence upon periodic-table architecture.

The principal distinction occurs between: (s, p) and (d, f) elements.

Elements belonging to the s- and p-blocks exhibit substantially larger radius ranges than elements belonging to the d- and f-blocks. This observation is particularly important because the analysis does not utilise any specific radius value.

Instead, it examines the spread between accepted radius definitions. Consequently, the periodic structure identified here cannot be attributed solely to the behaviour of a single radius system. Rather, it emerges from the relationships between multiple independent radius systems.

E9. Conclusions

The analysis presented in Appendix E establishes five principal observations.

1. Radius range width exhibits strong periodic-table dependence.
2. Group-level variation is substantial, with mean range widths varying from approximately 29 pm to 111 pm.
3. Periodic-table blocks form two statistically distinct populations.
4. The s- and p-blocks exhibit significantly larger radius ranges than the d- and f-blocks.
5. The observed separation is highly statistically significant and cannot reasonably be attributed to random measurement variation.

The results therefore demonstrate that atomic radius variation itself behaves as a structured periodic-table observable.

The spread between accepted radius definitions contains reproducible periodic information and is not adequately explained as arbitrary measurement disagreement alone.

This conclusion provides an important empirical foundation for subsequent interpretation of the physical origin of atomic radius variation.

Appendix F. Radius-Band Midpoint and Half-Range Amplitude Analysis

F1. Objective

Appendix E demonstrated that the total radius variation width,

$$\Delta R = R_{\max} - R_{\min}$$

is strongly dependent upon periodic-table group and block structure. Appendix F extends that analysis by decomposing the radius range into two quantities:

$$R_{mid} = \frac{R_{\max} + R_{\min}}{2}$$

and

$$A = \frac{R_{\max} - R_{\min}}{2}$$

where:

- R_{mid} is the midpoint of the observed radius band,
- A is the half-range amplitude,
- $R_{\max}$ is the largest accepted radius value for an element,
- $R_{\min}$ is the smallest accepted radius value for an element.

This decomposition tests whether the observed radius spread is better understood as a shifted radial band, an amplitude-like quantity, or both.

F2. Global Results

The analysis included: $N = 96$ elements with at least two valid radius measurements.

Metric	Mean	Median	SD	Min	Max
ΔR	75.16 pm	74.50 pm	24.90 pm	8.00 pm	124.00 pm
R_{mid}	160.56 pm	156.75 pm	36.24 pm	72.50 pm	304.00 pm
A	37.58 pm	37.25 pm	12.45 pm	4.00 pm	62.00 pm

The global mean half-range amplitude is therefore: $A \approx 38$ pm
This is a striking result because it suggests that the radius spread has a characteristic half-width scale.

F3. Half-Range Amplitude by Block

Block	N	Mean A	Median A	SD	Min	Max
s	14	45.32	47.00	12.07	23.00	62.00
p	30	44.73	44.50	9.49	25.50	59.50
d	29	31.98	33.50	8.82	12.00	44.00
f	23	30.59	31.50	12.86	4.00	59.50

The same population separation identified in Appendix E remains visible after converting ΔRinto half-range amplitude.

The s- and p-blocks form a high-amplitude population:

$$A_{s+p} = 44.92 \text{ pm}$$

The d- and f-blocks form a lower-amplitude population:

$$A_{d+f} = 31.37 \text{ pm}$$

F4. Hypothesis Testing of Half-Range Amplitude

The null hypothesis was:

$$H_0: \mu_s = \mu_p = \mu_d = \mu_f$$

The alternative hypothesis was that at least one block has a different mean half-range amplitude.

One-way ANOVA

$$F = 13.10$$
$$p = 3.39 \times 10^{-7}$$

Kruskal–Wallis non-parametric test

$$H = 28.98$$
$$p = 2.26 \times 10^{-6}$$

Both tests reject the null hypothesis. Therefore, half-range amplitude differs significantly by periodic-table block.

F5. Pairwise Block Comparisons

Comparison	p-value	Interpretation
s vs p	0.874	Not significantly different
s vs d	0.00147	**Significant**
s vs f	0.00148	**Significant**
p vs d	1.63×10^{-6}	**Significant**
p vs f	7.35×10^{-5}	**Significant**
d vs f	0.659	Not significantly different

This confirms the same two-population structure:

$$s \approx p$$
$$d \approx f$$
$$(s + p) \neq (d + f)$$

F6. Combined Population Test

The blocks were combined into two natural populations:

Population	Blocks	Mean A
Wide-amplitude population	s + p	44.92 pm
Narrow-amplitude population	d + f	31.37 pm

Welch unequal-variance test: $p = 8.64 \times 10^{-9}$
Mann–Whitney test: $p = 7.76 \times 10^{-8}$
Effect size: $d = 1.29$
This represents a large effect.

The amplitude separation between s/p and d/f elements is therefore both statistically significant and physically substantial.

F7. Radius-Band Midpoint by Block

Block	N	Mean R_{mid}	Median R_{mid}	SD	Min	Max
s	14	190.04	192.50	69.77	72.50	304.00
p	30	145.50	150.75	24.07	94.50	174.50
d	29	145.05	146.50	11.41	124.00	168.00
f	23	181.80	182.50	14.61	153.00	205.00

Midpoint behaviour is different from amplitude behaviour. The block midpoint distributions are not arranged in the same two-population pattern as the amplitude distributions. In particular:

- s-block and f-block elements have higher mean midpoint values,
- p-block and d-block elements have similar midpoint values,
- the combined s+p midpoint mean is statistically indistinguishable from d+f.

For combined populations:

$$R_{mid,s+p} = 159.67 \text{ pm}$$

$$R_{mid,d+f} = 161.31 \text{ pm}$$

Welch test: $p = 0.836$
Therefore, the s/p versus d/f distinction is not primarily a midpoint effect. It is primarily an amplitude effect.

F8. Correlation Analysis

Relationship	Pearson r	p-value
Amplitude vs Z	-0.311	0.00204
Midpoint vs Z	**0.490**	$\mathbf{4.02 \times 10^{-7}}$
Amplitude vs midpoint	-0.028	0.785
ΔR vs midpoint	**-0.028**	**0.785**

The most important result is:

$$r(A, R_{mid}) = -0.028$$

with:

$$p = 0.785$$

This means that half-range amplitude is essentially uncorrelated with the midpoint of the radius band.

This is a strong and unexpected result. It suggests that the width of the radius band is not simply a consequence of larger atoms having larger radii. Instead, the amplitude-like term appears to behave as an independent structural quantity.

F9. Interpretation

The analysis reveals three important features. First, the radius band has a characteristic global half-width:

$$A \approx 38 \text{ pm}$$

Second, half-range amplitude separates strongly by block:

$$A_{s+p} \approx 45 \text{ pm}$$

$$A_{d+f} \approx 31 \text{ pm}$$

Third, amplitude is not correlated with radius-band midpoint. This means that the observed radius spread is not merely a scaling artefact of atomic size. Larger midpoint radius does not automatically imply larger half-range amplitude. This result is consistent with the possibility that radius variation reflects an intrinsic radial shell-band property rather than simple proportional measurement uncertainty.

F10. Holland Atom Implication

Within the Holland Atom framework, this result is highly significant. If atomic shells are resonance-locked standing-mode structures, then each shell should possess:

- a mean radial position,
- a finite radial amplitude,
- an inner boundary,
- an outer boundary.

The present analysis maps naturally onto that structure: R_{mid} corresponds to the mean radial position of the observable shell band, while:

$$A = \frac{\Delta R}{2}$$

corresponds to a measurable half-amplitude of the radius band. The discovery that A is largely independent of R_{mid}, but strongly dependent upon block structure, suggests that the shell-band amplitude may be controlled by orbital-family or shell-geometry structure rather than simply by atomic size. This is consistent with the interpretation that s/p and d/f elements possess different radial-amplitude regimes.

F11. Conclusions

The radius-band decomposition produces a stronger result than the original ΔRanalysis alone.

The key findings are:

1. The observed radius range can be decomposed into a midpoint and half-range amplitude.
2. The global mean half-range amplitude is approximately 38 pm.
3. s- and p-block elements have a significantly larger half-range amplitude than d- and f-block elements.
4. The amplitude difference between s/p and d/f populations is highly statistically significant.
5. The effect size is large.
6. Radius-band amplitude is essentially uncorrelated with radius-band midpoint.
7. The s/p versus d/f separation is therefore an amplitude effect rather than merely a size-position effect.

These findings provide quantitative support for the interpretation that the spread between accepted radius definitions may represent a physically meaningful shell-band thickness.

Within the Holland Atom framework, this is consistent with the hypothesis that accepted atomic radius measurements sample different radial positions within a finite-amplitude resonance-locked shell structure.

Appendix G. Robustness Test: Systematic Removal of van der Waals Radius

G1. Objective

Appendices E and F demonstrated that the total radius range,

$$\Delta R = R_{\max} - R_{\min}$$

and the corresponding half-range amplitude,

$$A = \frac{\Delta R}{2}$$

exhibit strong block dependence across the periodic table.

However, because van der Waals radius is frequently the largest radius value within the dataset, it is necessary to test whether the observed signal is driven primarily by the inclusion of R_{vdW}.

The objective of Appendix G is therefore to repeat the radius-range and amplitude analysis after systematically excluding all van der Waals radius values.

This test determines whether the block-dependent radius-range structure survives when the dominant outer-boundary radius class is removed.

G2. Method

For each element, the radius range was recalculated using all available radius values except van der Waals radius.

The excluded dataset was: R_{vdW}, the retained datasets were:

- Empirical Atomic Radius
- Calculated Atomic Radius
- Cordero Covalent Radius (2008)
- Covalent Single-Bond Radius
- Covalent Triple-Bond Radius
- Metallic Radius

For each element with at least two remaining valid radius values, the following were recalculated:

$$\Delta R_{no-vdW} = R_{\max,no-vdW} - R_{\min,no-vdW}$$

$$A_{no-vdW} = \frac{\Delta R_{no-vdW}}{2}$$

$$R_{mid,no-vdW} = \frac{R_{\max,no-vdW} + R_{\min,no-vdW}}{2}$$

The resulting dataset contained: $N = 95$ elements.

G3. Global Effect of Removing van der Waals Radius

Metric	All Radii	Excluding vdW
N	96	95
Mean ΔR	75.16 pm	48.20 pm
Median ΔR	74.50 pm	51.00 pm
SD ΔR	24.90 pm	26.19 pm
Mean A	37.58 pm	24.10 pm
Median A	37.25 pm	25.50 pm

Removing van der Waals radius reduces the mean radius range by:

$$75.16 - 48.20 = 26.96 \text{ pm}$$

and reduces mean half-range amplitude by:

$$37.58 - 24.10 = 13.48 \text{ pm}$$

Thus, van der Waals radius contributes substantially to the total measured radius-band width.

This confirms that the original amplitude result is partly driven by the separation between van der Waals radii and the bonding/empirical/covalent radius family.

G4. Block Dependence After Removing van der Waals Radius

Block	N	Mean ΔR_{no-vdW}	Median ΔR_{no-vdW}	SD
s	13	49.00 pm	47.00 pm	26.57 pm
p	30	23.37 pm	20.50 pm	10.58 pm
d	29	63.24 pm	67.00 pm	18.19 pm
f	23	61.17 pm	63.00 pm	25.71 pm

The block structure survives, but the ordering changes.

With van der Waals radius included, the broad population was:

$$s + p > d + f$$

After removing van der Waals radius, the pattern becomes:

$$d + f > s + p$$

This is a major finding.

It indicates that the full-range shell-width result is not simply reproduced unchanged after removing R_{vdW}. Instead, van der Waals radius is responsible for much of the large s/p-band expansion observed in the original analysis.

However, the block dependence itself does not disappear.

G5. Statistical Significance Without van der Waals Radius

The null hypothesis is:

$$H_0: \mu_s = \mu_p = \mu_d = \mu_f$$

for ΔR_{no-vdW}.

One-way ANOVA

$$F = 24.71$$
$$p = 8.74 \times 10^{-12}$$

Kruskal–Wallis Test

$$H = 43.87$$
$$p = 1.61 \times 10^{-9}$$

The null hypothesis is strongly rejected.

Therefore, even after removing van der Waals radius, radius-range width remains highly dependent upon periodic-table block.

G6. Combined Block Population Test

The original all-radii analysis found:

$$A_{s+p} = 44.92 \text{ pm}$$
$$A_{d+f} = 31.37 \text{ pm}$$

After removing van der Waals radius:

$$A_{s+p,no-vdW} = 15.56 \text{ pm}$$
$$A_{d+f,no-vdW} = 31.16 \text{ pm}$$

Thus, removal of R_{vdW} reverses the population ordering.

Welch Test

$$p = 1.65 \times 10^{-10}$$

Mann–Whitney Test

$$p = 4.85 \times 10^{-9}$$

Effect Size

$$d = -1.48$$

The negative sign reflects the reversal of direction: $d + f$ now has the larger internal radius spread.

The effect remains large.

G7. Amplitude–Midpoint Relationship After Removing van der Waals Radius

The all-radius analysis found:

$$r(A, R_{mid}) = -0.028$$

$$p = 0.785$$

indicating that half-range amplitude was essentially independent of atomic size.

After removing van der Waals radius, the relationship becomes:

$$r(A_{no-vdW}, R_{mid,no-vdW}) = 0.564$$

$$p = 2.57 \times 10^{-9}$$

This is a major change.

Without van der Waals radius, the internal radius spread becomes moderately and significantly correlated with radius-band midpoint. Therefore, the striking amplitude–midpoint independence observed in Appendix F depends upon inclusion of the outer interaction boundary represented by van der Waals radius.

This does not invalidate the result, but it clarifies its meaning. The independence result applies to the full observable radius band, including the outer non-bonded interaction boundary. It does not apply to the reduced internal/bonding radius family alone.

G8. Interpretation

The systematic removal of van der Waals radius produces three key findings.

First, the total radius range decreases substantially. This confirms that van der Waals radius is a major contributor to the full shell-band width.

Second, block dependence survives removal of van der Waals radius. Therefore, the periodic structure is not an artefact caused solely by R_{vdW}.

Third, the direction of block separation reverses. With R_{vdW} included, s- and p-block elements exhibit the larger full shell-band amplitude. Without R_{vdW}, d- and f-block elements exhibit the larger internal radius-family spread.

This suggests that the full shell-band structure contains at least two components:

1. An outer-boundary component dominated by van der Waals interaction behaviour.
2. An internal radius-family component dominated by covalent, empirical, calculated, metallic, and bonding-related radii.

The original standing-mode amplitude interpretation should therefore be refined.

The full radius band is not a single homogeneous measurement. It is a composite observable spanning internal, bonding, empirical, metallic, and outer interaction regions.

G9. Implications for the Standing-Mode Amplitude Hypothesis

The van der Waals removal analysis strengthens the empirical case while requiring refinement of the physical interpretation. The data do not support a simplistic claim that:

$$A = \frac{R_{vdW} - R_{min}}{2}$$

is automatically a symmetric standing-wave amplitude. Rather, the data support a more nuanced conclusion: the full observable shell band contains multiple radial regimes.

The van der Waals radius appears to define, or strongly approximate, the outer interaction boundary of that band. The smaller radii define internal, bonding, empirical, or structural regions within the same band. The full-band amplitude therefore reflects the complete observable radial envelope, while the no-vdW analysis reflects the internal radius-family spread. This distinction is important and should be explicitly stated in Section 5 and Figure 1. The refined hypothesis becomes:

The standing-mode shell may possess a finite radial envelope. Different radius families probe different regions of that envelope. The van der Waals radius is not interchangeable with bonding radii, but may represent the outer interaction limit of the shell.

This framing directly addresses the reviewer's criticism.

G10. Conclusions

The robustness test demonstrates that the radius-range signal survives systematic removal of van der Waals radius, but its interpretation changes.

The principal conclusions are:

1. van der Waals radius contributes substantially to the full observed radius-band width.
2. Removing van der Waals radius reduces mean ΔR from 75.16 pm to 48.20 pm.
3. Removing van der Waals radius reduces mean amplitude from 37.58 pm to 24.10 pm.
4. Block dependence remains highly statistically significant after removing van der Waals radius.
5. The direction of block separation reverses: with van der Waals included, s/p elements have the larger full-band amplitude; without van der Waals, d/f elements have the larger internal-family amplitude.
6. The amplitude–midpoint independence result applies to the full observable shell band but not to the internal radius family alone.

7. The data therefore support a two-component interpretation of atomic radius variation: an internal radius-family component and an outer interaction-boundary component.

Overall, this analysis does not invalidate the standing-mode amplitude hypothesis. It refines it.

The full radius-band amplitude should not be described as a simple symmetric shell thickness derived from equivalent radius classes. Instead, it should be described as the observable radial envelope produced when multiple valid radius families sample different regions of atomic structure, with van der Waals radius acting as the dominant outer-boundary constraint.

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